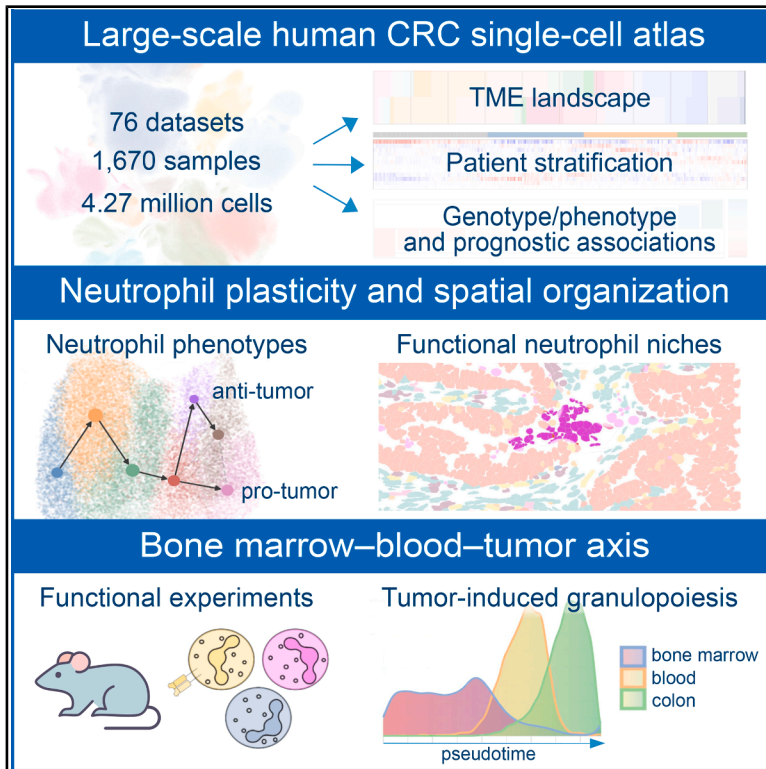


Single-cell integration and multi-modal profiling reveals phenotypes and spatial organization of neutrophils in colorectal cancer

Graphical abstract



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In brief

Marteau et al. integrate colorectal cancer single-cell datasets, classify immune phenotypes, and identify myeloid gene expression programs. Characterization of neutrophils reveals pro- and antitumorigenic properties, spatial neutrophil niches, and tumor-derived reprogramming of neutrophils in the bone marrow.

Highlights

- Generation of a large-scale CRC atlas integrating 4.27 million cells from 1,670 samples
- Characterization of pro-tumorigenic and anti-tumorigenic neutrophils in patients with CRC
- Spatial multimodal single-cell profiling reveals functional neutrophil niches
- Orthotopic mouse model indicates tumor-induced granulopoiesis



Article

Single-cell integration and multi-modal profiling reveals phenotypes and spatial organization of neutrophils in colorectal cancer

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SUMMARY

The immune composition of the tumor microenvironment has a major impact on the therapy response in patients with colorectal cancer. Here, we built an atlas with 4.27 million single cells from 1,670 patient samples and complemented it with single-cell profiles from 266 patients, including cells with low mRNA content, spatial transcriptomics from 3.7 million cells, and protein profiles from 0.7 million cells. The analysis of the atlas allows tumor classification into immune desert, B cell enriched, T cell enriched, and myeloid cell enriched immune phenotypes. Within the myeloid compartment, we identify consensus myeloid gene expression programs with four immunomodulatory programs, and uncover a subpopulation of neutrophils with antigen-presenting properties. Moreover, functional experiments using patient-derived organoids show KRAS-dependent pro-tumorigenic polarization of neutrophils. Further, spatial multimodal single-cell profiling reveals niches with IL-1 signaling-based neutrophil-fibroblast interaction. Finally, using an orthotopic mouse model, we show that cancer-derived signals modify neutrophil production in the bone marrow.



INTRODUCTION

Colorectal cancer (CRC) is the most prevalent gastrointestinal malignancy and represents the second leading cause of cancer-related mortality worldwide according to the GLOBOCAN data.¹ Despite the improvement of early detection screening policies and the development of many treatments, the median overall survival (OS) for patients with metastatic CRC (mCRC) is ~30 months.² Advancements in targeted therapy for mCRC have been limited, especially when contrasted with the significant progress achieved in other solid tumors such as non-small cell lung cancer.³ Moreover, CRC appears to exhibit resistance to immune checkpoint inhibitor (ICI) therapy, with the notable exception of approximately 4% of metastatic cases characterized by a damaged mismatch repair system (dMMR) or/and high microsatellite instability (MSI-H).^{4,5} This is somehow paradoxical since CRCs are known to be under immunological control.⁶ Thus, there is an urgent need to explore new immunomodulatory strategies for the vast majority of patients with CRC with microsatellite stable (MSS) tumors.

The advent of single-cell RNA sequencing (scRNA-seq) technologies opened fundamental new avenues to dissect cancer heterogeneity. We recently developed a computational approach to dissect the cellular diversity at high resolution by integrating expression profiles from multiple studies using artificial intelligence (AI) methods.⁷ A similar effort recently carried out in the context of CRC⁸ using the 10x Genomics platform is lacking neutrophils, the most abundant circulating cells, which are emerging as important regulators of cancer.^{9–12} Neutrophils are highly abundant in intratumoral as well as stromal regions in CRC,¹⁰ co-localize with CD8⁺ T cells,¹¹ show different pro-tumor and anti-tumor functions,¹² and can acquire an anti-tumor phenotype in response to immunotherapy.¹³ Hence, deep characterization of neutrophils in patients with CRC would provide novel insights into the myeloid compartment and the myeloid-lymphoid cell-cell interactions.

Here, we integrated 76 CRC single-cell RNA sequencing datasets spanning 4.27 million cells from 650 patients and 1,670 samples. We complemented the atlas by profiling single cells from four cohorts ($n = 266$ patients), including: 1) single-cell profiling of samples using an optimized protocol that efficiently recovers low mRNA-content cells; and 2) spatial single-cell transcriptional profiling of 3.7 million cells as well as spatial single-cell protein profiling of 715,000 cells (Figure S1A). We then used this big data resource to determine immune phenotypes, identify consensus myeloid expression programs, and characterize neutrophils, including the quantification of spatial niches. Additionally, we charted the tumor-bone marrow axis at the single-cell level using an orthotopic mouse model and provided functional data using this model and patient-derived organoids.

RESULTS

Generation of a large-scale colorectal cancer single-cell atlas

Using our approach for the integration of single-cell data,⁷ we compiled CRC related scRNA-seq data from 48 studies and 76 datasets comprising 4.27 million cells from 650 patients and 1,670 samples (Figure 1A). This CRC single-cell atlas integrates

data from publicly available studies, complemented with meta-data curation, and encompasses the full spectrum of disease progression, from normal colon to polyps, primary tumors, and metastases, covering both early and advanced stages of CRC (see STAR Methods, Figures 1B and 1C). Study characteristics are summarized in Table S1. In total, the atlas includes transcriptomic data from 487 patients with CRC (median: 58 years [22–91]), 99 patients with polyps (median: 60 years [40–75]), and 64 control individuals (median: 57 years [23–79]). Of the 487 patients with CRC, 254 had colon cancer (COAD), 63 had rectal cancer (READ), and 170 were not otherwise specified. Of these, 89 were MSI-H, 268 MSS, and 130 had an unknown microsatellite status. In terms of disease stage, 41 patients were classified as TNM stage I, 121 as stage II, 171 as stage III, and 110 as stage IV, while the stage was not reported for 44 patients. Samples include tissue of the primary tumor ($n = 674$) or CRC liver metastasis ($n = 175$) that were obtained by surgical resection. Among the control samples, 284 were derived from distant non-malignant tissue of patients with colon tumor, of which 220 had a patient-matched tumor sample, 186 non-tumor control samples, as well as 116 polyps and 90 matched normal samples obtained from patients undergoing colonoscopy. Further, 103 blood samples of patients with CRC, as well as 24 samples derived from tumor affected lymph nodes, with another 18 non-tumor affected lymph node samples, were included.

Neutrophils are underrepresented in most scRNA-seq datasets generated with 10x Genomics technology.¹⁴ We therefore generated scRNA-seq data from patients with CRC from cohort I ($n = 12$) using the BD Rhapsody platform and our optimized protocol¹⁵ that allows capture of cells with very low mRNA content.¹⁴ The data were then integrated into the final high-resolution single-cell atlas. The atlas integrates 4,264,946 single cells, which are annotated to 12 coarse cell types and 61 major cell subtypes or cell states, including 1,146,196 epithelial cells, 1,552,108 T/NK cells, 804,042 B/Plasma cells, 385,589 myeloid cells, and 33,9767 stromal and endothelial cells (Figures 1D and S1B–S1G). Cancer cells (600,740 cells, from primary tumor and metastatic tissue) in the atlas showed high heterogeneity in their transcriptomic profiles (Figure 1D). Due to the large patient cohort, we were able to apply high-resolution cancer cell classification based on their specific marker gene expression signatures. We divided the following main clusters: 1) colonocyte-like (*FABP1*, *CEACAM7*, *ANPEP*, *SI*), 2) BEST4 (*BEST4*, *OTOP2*, *SPIB*), 3) Goblet-like (*MUC2*, *FCGBP*, *ZG16*, and *ATOH1*), 4) Crypt-like (*LGR5*, *SMOC2*), and 5) transit-amplifying-like (*TOP2A*, *UBE2C*, *PCLAF*, *HELLS*, *TK1*) (Figure 1D). There are highly mitotic/proliferative clusters (*LGR5*, *TOP2A*, and *MKI67*) of both COAD and READ that resemble highly aggressive and invasive cancer cells. In addition, *EPCAM* expressing cells in the blood of patients with CRC were annotated as circulating cancer cells.

With a total of 2,741,739 immune cells, this atlas enables the high-resolution classification of immune cell types across both innate and adaptive subsets (Figures 1E and S1B–S1D). These include T cells - CD4 (*CD4*), CD8 (*CD8A*), T regulatory cells (*FOXP3*), and NK cells (*NG2*, *TYROBP*). B cells are represented as naive (*IGHD*, *TCL1A*), memory (*CD27*, *CD44*), and activated (*CD69*, *CD83*) subsets, while plasma cells (*JCHAIN*) express immunoglobulins such as IgA (*IGHA1*), IgG (*IGHG1*), and IgM

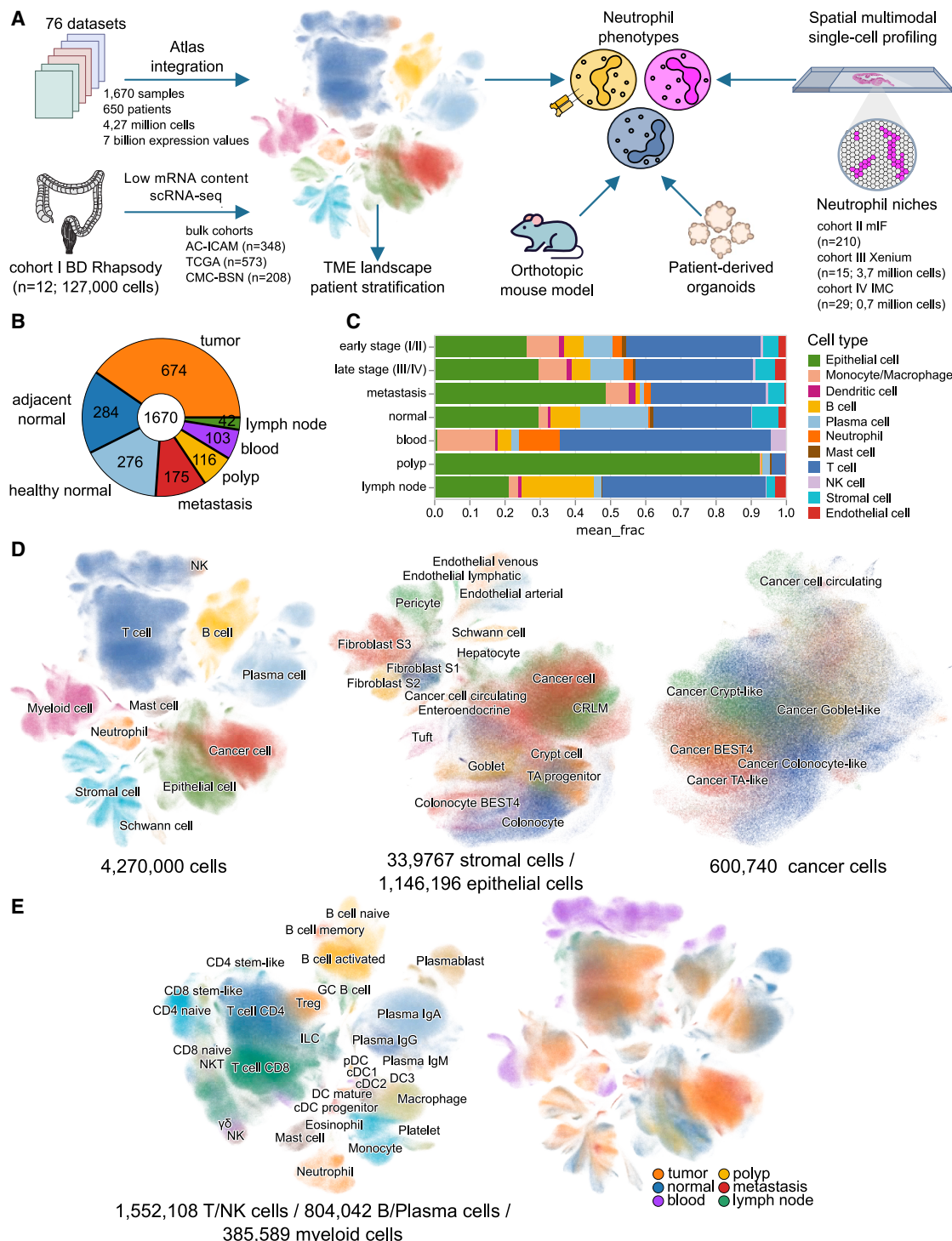


Figure 1. Schematic outline of the overall concept of the study, including publicly available and generated datasets

(A) Summary of the data integration and analysis workflow.

(B) Number of samples for each type.

(C) Overview of the cell fractions in different tissues.

(D) Overview of the CRC atlas and the epithelial and stromal/endothelial components depicted as UMAP plots.

(E) Overview of the CRC atlas sample types and the immune components depicted as UMAP plots. See also [Figure S1](#) and [Table S1](#).

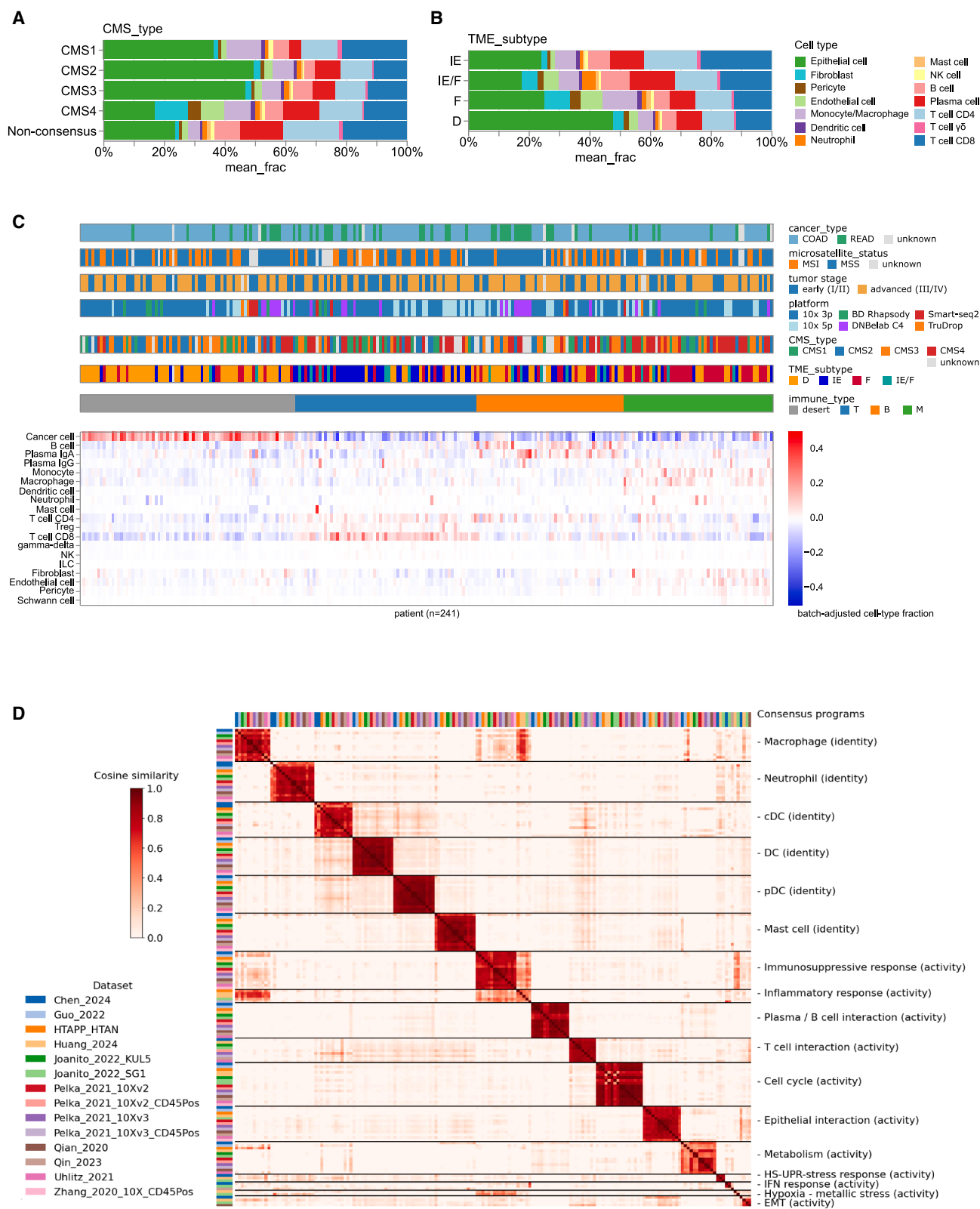


Figure 2. CRC patient stratification based on tumor immune phenotypes

(A and B) Cellular composition of non-treated patients classified based on bulk transcriptomic data according to the (A) molecular subtypes (CMS1-4) and (B) TME subtypes: immune-enriched/fibrotic (IE/F), immune enriched (IE), fibrotic (F), and depleted (D), estimated cellular composition of the TME using metagene scores and deconvolution methods.

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(*IGHM*). The myeloid lineage encompasses monocytes (*VCAN*, *CD14*), macrophages (*C1QA*, *C1QB*), mast cells (*TPSB2*, *CPA3*), dendritic cells, including conventional DC type 1 (*cDC1*, *CLEC9A*, *XCR1*) and cDC2 (*CD1C*, *CLEC10A*), and plasmacytoid DC (*pDC*, *IL3RA*, *GZMB*), as well as eosinophils (*CLC*, *CCR3*) and neutrophils (*CXCR2*, *FCGR3B*). Specialized populations such as gamma-delta T cells (*TRDV1*, *TRGC1*) and innate lymphoid cells (ILCs; *IL4I1*, *KIT*) are also depicted, along with immune cell developmental stages, such as progenitors (*CD24*, *AXL*) and stem-like cells (*LEF1*), and cycling states (*MKI67*, *CDK1*).

TME characterization at the single-cell level enables the stratification of tumors into distinct immune phenotypes

Two cancer classification subtypes based on bulk transcriptomic data were previously reported: the consensus molecular subtypes CMS1-4¹⁶ and the TME subtypes: immune-enriched/fibrotic (IE/F), immune enriched (IE), fibrotic (F), and depleted (D). These classifications estimated the cellular composition of the TME using metagene scores and deconvolution methods.¹⁷ We used the CRC atlas to provide a high-resolution view of the cellular composition of the TME. Fractions of the major cell types, epithelial, endothelial, fibroblasts, and immune cells, corresponded to the original classifications (Figures 2A and 2B). Within the immune compartments, the fractions of innate and adaptive immune cell types were highly diverse. For example, cellular fractions of monocytes and T cells were decreasing from CMS1-CMS4, whereas B cells/plasma cells were increasing (Figures S2A and S2B). In order to characterize these immune phenotypes, we performed Leiden clustering of batch-corrected cell-type fractions, and revealed four clusters (Figures 2C and S2C): 1) immune deserted tumors (ID; i.e., no significant immune cell infiltration); 2) the subtype of tumors with T cell dominance (T; CD8⁺, CD4⁺, T regulatory cells); 3) the subtype of tumors with B cell dominance (B; B cell, plasma IgA and IgG cell) and 4) the subtype of tumors with myeloid dominance (M; macrophage/monocyte). Patients in the T strata exhibited significantly higher odds of MSI (adjusted OR 2.75, 95% CI 1.03–7.33; *p* = 0.043).

Identification of consensus myeloid gene expression programs

Given the role of myeloid cells in creating immunosuppressive TME, we aimed to identify consensus myeloid gene expression programs and used non-negative matrix factorization (NMF).¹⁸ Noteworthy, gene programs identified using NMF represent fundamental components of myeloid cells that cannot be distinguished by conventional clustering.¹⁸ The identified consensus myeloid expression programs included cell-activity programs and cell-identity programs (Figures 2D and S2D). The cell-activity consensus programs included cell cycle, epithelial cell interaction, metabolism, as well as environmental cues, such as the heat shock/unfolded protein response, interferon response, and hypoxia (Figure 2D). Importantly, we also identified four immunomodulatory programs: immunosuppressive response,

inflammatory response, as well as the consensus programs B cell/plasma cell interaction and T cell interaction. These immunomodulatory programs suggest intricate innate-adaptive immune cell crosstalk in the TME. The cell-identity programs included DC, pDC, cDC, macrophages, mast cells, and neutrophils. The consensus program neutrophils were characterized by inflammatory genes such as *IL1A* and *IL1B*, together with tumor-associated neutrophil (TAN) markers like *OLR1*, *VEGFA*, and *PLAU*⁷ (Table S2), implying functional diversity. Strikingly, usage of consensus gene expression programs also revealed the T cell interaction program in neutrophils (Figure S2D), further supporting the dual role of neutrophils in tumor promotion and suppression.

Integration of bulk RNA-seq data reveals genotype-immune phenotype associations

The majority of the scRNA-seq studies lack both comprehensive cancer genotype information and survival data. In contrast, the TCGA¹⁹ and the more recent AC-ICAM²⁰ and CMC-BSN²¹ cohorts include this information, alongside bulk RNA-sequencing data. Building on our previous work that identified genomic features influencing immune phenotypes using bulk RNA-seq data,²² we leveraged the high resolution of the single-cell CRC atlas for a more detailed analysis of these determinants. We first developed Shears, a computational method to address major limitations of existing algorithms for associating single cells from scRNA-seq datasets with bulk RNA-seq phenotypic or clinical information, such as computational inefficiency on large datasets, and the inability to account for covariates and biological replicates (STAR Methods). Briefly, Shears utilizes a weighted single-cell matrix rather than the correlation matrix to account for the relative contribution of each cell, and computes the importance of each cell for a given phenotype, fitting a linear model with covariates (STAR Methods). We then combined bulk RNA-seq data from the TCGA (*n* = 573), the AC-ICAM (*n* = 348), and the CMC-BSN (*n* = 208) cohorts and used Shears to assess the relationships between survival outcomes as well as specific genotypes with atlas-derived cell-type transcriptomic signatures (Figures 3A–3D and S3A–S3D). Overall, the presence of B cells, plasma cells, and eosinophils, as well as several T cell subtypes including CD8⁺ T cells, CD4⁺ T cells, and $\gamma\delta$ T cells, was associated with improved survival in patients with CRC (Figures 3A and S3A). Conversely, patients with CRC with tumors enriched with endothelial cells, stromal cells, and neutrophils were associated with worse survival outcomes (Figures 3A and S3A). Tumors from patients with MSI CRC were infiltrated with adaptive and innate immune cells, including T cells and neutrophils, compared to tumors from patients with MSS CRC (Figures 3B and S3B). Distinct tumor genotypes such as *BRAF* and *KRAS* mutated tumors, were associated with different immune infiltrates (Figures 3C, 3D, S3C, and S3D). For example, tumors from patients with CRC with *KRAS* mutations had increased infiltrations of neutrophils, whereas the same cell types were depleted in tumors from patients with CRC with *BRAF* mutations (Figures 3C, 3D, S3C, and S3D).

(C) Patient characteristics and stratification of the tumor immune phenotypes.

(D) Heatmap depicts similarity of the gene spectra scores of each program in the 14 discovery datasets identified using non-negative matrix factorization (NMF). Consensus programs created from their average scores are identified with black horizontal lines. See also Figure S2 and Table S2.

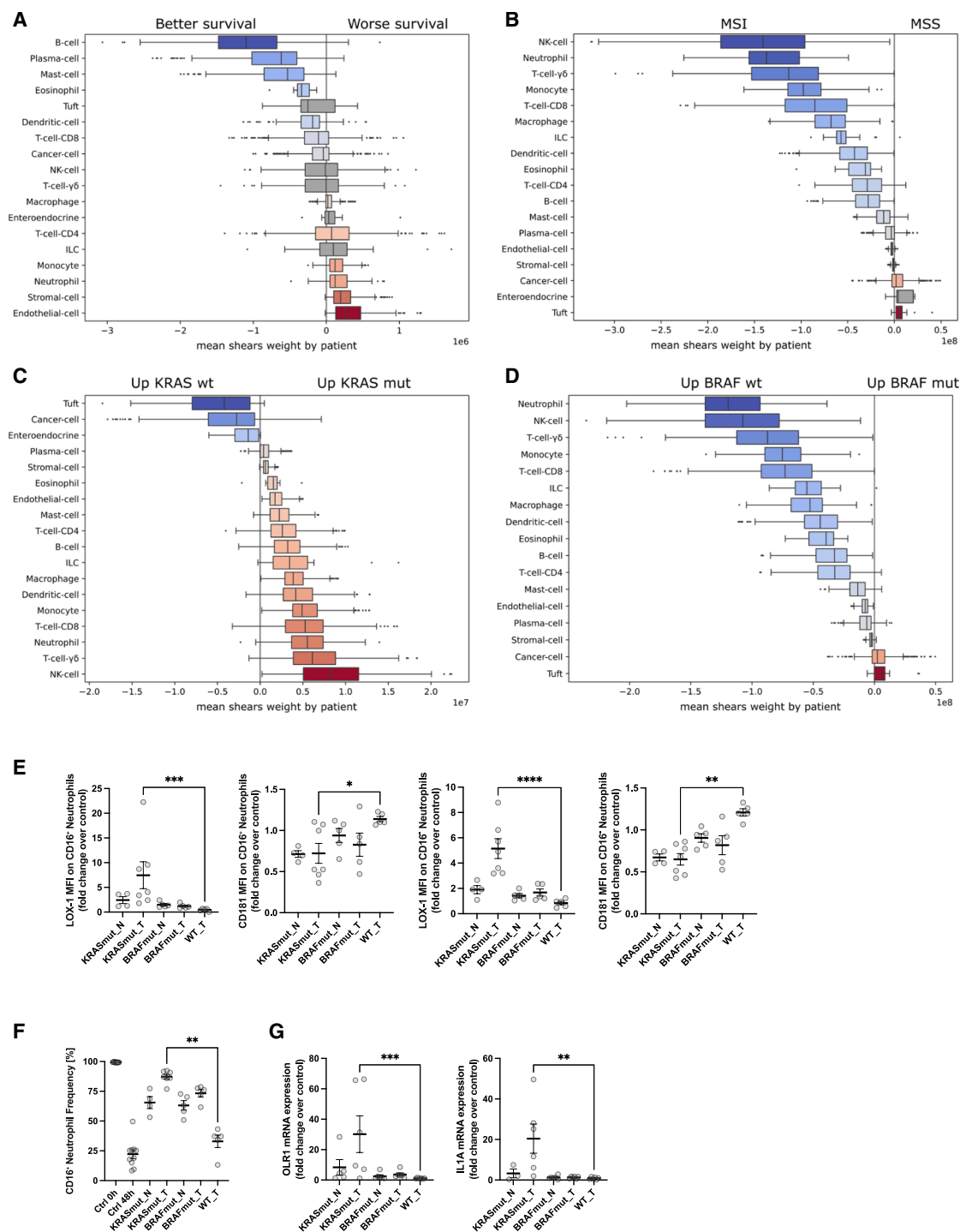


Figure 3. Associations of cellular composition, survival, and genotypes

(A–D) Shears analysis of bulk RNA-seq data from the AC-ICAM, TCGA, and the CMC-BSN cohorts. Boxplots show the mean Shears' coefficient per patient and cell type (≥ 20 cells), weighted by $-\log_{10}(p\text{-value})$. Boxes indicate the interquartile range (IQR) with the median as the central line, whiskers extend to $1.5 \times \text{IQR}$; colored boxplots denote significant associations, and gray boxplots denote non-significant ones. Significance was assessed by the Wilcoxon signed-rank test with FDR correction ($\text{FDR} < 0.05$). UMAP plots indicate the position of cells positively (red) or negatively (blue) associated with mutation or better survival. (A) Association of cellular composition with overall survival ($n = 968$). (B) Cellular composition of patients with MSI ($n = 151$) vs. MSS ($n = 821$). (C) Cellular composition of KRAS mutated (383) vs. wildtype tumors ($n = 589$). (D) Cellular composition of BRAF mutated ($n = 122$) vs. wildtype tumors ($n = 653$). (E) Flow cytometry quantification of LOX-1 and CD181 expression on neutrophils after 48 h exposure to pre-conditioned media from CRC organoids. KRAS/BRAFmut_N and KRAS/BRAFmut_T: normal tissue and tumor tissue from patients with respective mutations. (F) Frequency of CD16⁺ neutrophils after 48 h exposure to

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To explore mechanisms underlying these genotype-immune phenotype associations, we performed functional co-culture experiments with primary neutrophils from healthy donors and conditioned media from patient-derived CRC organoids.²³ Exposure to KRAS-mutant tumor organoid media led to a significant increase in Lectin-type oxidized LDL receptor 1 (LOX-1; encoded by *OLR1*) and a reduction in CD181 surface expression on CD16⁺ as well as CD16[−] neutrophils (Figure 3E), both established markers of a TAN phenotype.⁷ In addition, KRAS-mutant organoid media preserved the proportion of CD16⁺ neutrophils, a marker of mature neutrophils,²⁴ and enhanced neutrophil survival (Figures 3F and S3E). At the transcriptional level, *OLR1* and *IL1A* were upregulated in neutrophils exposed to KRAS-mutant tumor organoid media, consistent with functional reprogramming (Figure 3G). In contrast, BRAF-mutant tumor organoid media induced only modest effects on TAN marker expression, while wild-type tumor organoid media consistently showed the weakest impact, even compared to KRAS/BRAF-mutant normal-adjacent tissue organoid media (Figures 3E–3G). These findings were confirmed using CRC cell lines, where KRAS-mutant lines (LoVo, HCT116) preserved the proportion of CD16⁺ neutrophils and promoted both neutrophil survival and *OLR1* induction, whereas BRAF-mutant (HT-29) and KRAS/BRAF wild-type (CaCo2) lines had minimal effects (Figure S3E). Functionally, neutrophils reprogrammed by KRAS-mutant tumor organoid media exerted an inhibitory effect on CD8⁺ T cell activation, reflected by reduced CD69 expression, while CD4⁺ T cells remained unaffected (Figure S3E). Together, these results support a KRAS-dependent mechanism promoting neutrophil survival and TAN-like polarization, providing a functional link to the neutrophil-enriched TME characteristic of KRAS-mutant CRC.

Circulating and tumor-associated neutrophils have diverse and plastic phenotypes

These dual roles of neutrophils in CRC and the lack of neutrophil data in past studies inspired us to use the single-cell atlas to carry out the comprehensive characterization of the data we generated using the BD Rhapsody platform. The obtained scRNA-seq dataset comprises 126,991 single cells categorized into 23 main cell types based on specific marker genes, including 55,245 epithelial/tumor cells, 4,428 stromal/endothelial cells, and 65,782 immune cells, including 30,924 neutrophils (Figure 4A). Compositional analysis using scRNA-seq and flow cytometry showed high concordance when comparing blood, normal, and tumor tissue samples (Figure 4B). As expected, neutrophils were the most abundant leukocyte subtype in blood samples (60–70%), followed by T lymphocytes and monocytes (Figure 4B). Normal-adjacent tissue was dominated by epithelial cells, while tumor tissue contained a higher proportion of immune cells. Notably, neutrophils comprised ~20% of all cells and ~40% of leukocytes in tumor tissue but were present only in small numbers in normal-adjacent tissue, which was confirmed using flow cytometry (Figure 4C). Within the granulocyte lineage, neutrophils were the most abundant cells in the tumor and in blood (95.2% and 96.1%, respectively), whereas in normal adjacent tissue, the fractions for neutrophils (36.8%), eosinophils (35.6%), and mast cells (27.6%) were similar.

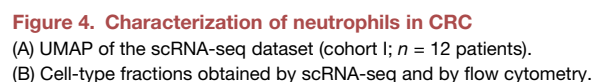
We observed distinct transcriptomic differences between blood neutrophils versus TAN (Figure 4D) with 2,214 differentially expressed genes ($p_{adj} < 0.05$; Figure 4E). We evaluated previously described marker genes distinguishing TAN from normal tissue neutrophils (NAN) across cancers, including *OLR1*, *VEGFA*, *CXCR2*, and *PLAU*.²⁵ TAN markers (*OLR1*, *PLAU*, and *VEGFA*) were highly upregulated in TAN of patients with CRC, whereas *CXCR2*, a marker of mature neutrophils, was highly expressed in blood neutrophils but downregulated in TAN (Figure 4F). Spatial analysis of LOX-1 expression using multiplex immunofluorescence (IF) staining demonstrated the presence of segregated LOX-1⁺ neutrophil cell clusters exclusively in CRC tumor tissue, but not in normal-adjacent colonic tissue (Figures 4G and S4A). Of note, LOX-1 has previously been identified as a marker for TAN in lung cancer,⁷ indicating LOX-1 as a putative tumor-agnostic TAN marker.

Sub-clustering of neutrophils revealed three distinct blood neutrophil clusters (BN1–BN3) and four TAN clusters (TAN1–TAN4) (Figure 4H). Trajectory analysis indicated a transition from BN1 to BN3 and subsequently to TAN1 (Figure 4I), demonstrating that neutrophil migration from the bloodstream into tissue is accompanied by distinct transcriptomic changes. From TAN1, neutrophils diverged into two trajectories: one leading to a putative pro-tumor TAN phenotype (TAN4) characterized by the high expression of *IL1A* and *IL1B*. Of note, *IL1A* is a key driver of emergency myelopoiesis that fuels inflammation and immunosuppression in the TME.²⁶ The other trajectory was leading toward a phenotype defined by the high expression of antigen-presentation-associated markers, such as *HLA-DRA* (TAN2/TAN3) (Figure 4J). We then validated HLA-DR protein expression on neutrophils using flow cytometry and confirmed that the expression was markedly higher in TAN compared to both blood neutrophils and NAN (Figure 4K). Neutrophils with antigen-presenting capabilities, referred to as hybrid neutrophils,²⁵ have previously been identified in early-stage lung cancer and are associated with an anti-tumor phenotype.^{7,27} The analyses of the expression data showed that in these neutrophil subtypes several HLA-DR genes were upregulated, whereas calprotectin (S100A8/S100A9) was downregulated (Figure S4B). Analysis of the cell-cell communication using ligand-receptor methods (LIANA)²⁸ indicated HLA-DR⁺ neutrophil-T cell interactions (Figures S4C and S4D).

To investigate the prognostic value of HLA-DRA⁺ neutrophils, we used tissue microarrays constructed from an independent patient cohort (cohort II, $n = 210$) and multiplexed IF stainings (Figures S4E and S4F). HLA-DRA⁺ neutrophils were detectable in both the tumor core and the invasive margin as clusters of cells, albeit at slightly different fractions (6.8% and 7.1% of the total neutrophils, respectively, Figure S4G). Patients with tumors with high densities of HLA-DRA⁺ neutrophils in the tumor core

pre-conditioned media from the CRC organoids. (G) *OLR1* and *IL1A* mRNA expression in neutrophils after 24 h exposure to pre-conditioned media from the CRC organoids. RT-qPCR values were normalized to GAPDH.

(E–G) Each dot represents the mean value from an individual healthy blood donor ($n = 3–10$). Data are shown as means $\pm$ SEM. Statistical significance was assessed using the Kruskal-Wallis test with Dunn's post hoc correction for multiple comparisons. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. See also Figure S3.



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had better outcomes compared to patients with low densities (Figure 4L). This effect was stronger for early stages compared to more advanced stages (Figure S4H). Additionally, the presence of HLA-DRA⁺ neutrophils in the tumor margin had an opposite effect in both early and late stage tumors (Figure S4H), suggesting that the localization of these cell types plays a major role in the control of tumor progression.

Spatial single-cell multi-modal profiling reveals functional neutrophil niches

A peculiar observation of the multiplexed IF stainings was that neutrophils in the TME occupy spatially distinct areas or patches (Figures 4G and S4D), suggesting the existence of specific cellular compositions and discrete histological properties: so-called cellular neighborhoods²⁹ or cell niches. However, due to the limited number of features in the used multiplexed IF assay, it was not possible to deduce cell-cell interactions from these spatial profiling datasets. In order to quantify spatial niches in general, and specifically neutrophil niches, we therefore applied the spatial transcriptomic technique with 380 genes and cellular resolution (10x Genomics Xenium platform), and used FFPE tumor samples from patients with CRC (cohort III, $n = 15$). In total, 3.7 million cells were profiled, resulting in 217 million expression values. The spatial single-cell transcriptomic profiling revealed distinct spatial architecture and cell type composition in both, center of the tumor and the invasive margin (Figure 5A). Whereas normal adjacent tissue sections recapitulated the known morphology of healthy colon epithelium, tumor cores and invasive margins were more heterogeneous in their spatial architectures and immune infiltration patterns. The cellular composition of the samples was heterogeneous, and neutrophils were detectable in all samples (Figure 5A). Neutrophils were spatially localized as aggregates or clusters in both the tumor core and the invasive margin (Figure 5B). This spatial organization of neutrophils was confirmed using H&E stainings from the same slide as well as immunohistochemistry (IHC) stainings of consecutive FFPE slices confirmed by two board certified pathologists (S.O. and M.G.) (Figure S5A).

We then aimed to quantitatively define cellular niches from their biological functions and used a graph deep-learning method that utilizes existing knowledge of inter- and intracellular interaction pathways to identify such functional cellular niches (Niche Compass)³⁰ (Figure S5B). Niches were able to differentiate between tumor and stroma regions. At a higher resolution, multiple cancer niches appeared, which we further summarized

based on their cell composition into cancer, fibroblast-enriched cancer, and myeloid-enriched cancer niches (Figures 5C and S5B). Neutrophils were mainly present in the myeloid-enriched niche and in the neutrophil niche. In the neutrophil niche, most of the neutrophils were organized into clusters (Figures S5C and S5D) and had a pronouncedly increased expression of *IL1B* (Figure 5D). Fibroblasts and, to a lesser extent, also the cancer cells in this niche upregulated the chemokines *CXCL5*, *CXCL3*, and *CXCL1* (Figures 5E and S5E). Further analysis of the main cell types within the neutrophil niche revealed that the receptor for *IL1B*, *IL1R1*, was specifically expressed in fibroblasts (Figure 5F). Cell-cell-interaction analysis (LIANA) of cells in the neutrophil niche further indicated that the increased expression of *CXCL5*, *CXCL3*, and *CXCL1* by fibroblasts recruits neutrophils via *CXCR1* and *CXCR2* (Figure S5F). Of note, it was previously reported that IL-1 signaling in tumor cells polarizes cancer-associated fibroblasts toward an inflammatory phenotype, which mediates resistance to neoadjuvant therapy in rectal cancer.³¹ A phase I clinical trial inhibiting IL-1 with recombinant IL-1RA (anakinra) in combination with chemoradiotherapy in patients with rectal cancer was recently completed (ClinicalTrials.gov Identifier: NCT04942626). Here, we show that neutrophils, rather than cancer cells, express *IL1B* (Figures 5D, S5E, and S5F) and thereby polarize cancer-associated fibroblasts.

To validate the findings based on mRNA expression in single cells, we then used an orthogonal protein-based method (imaging mass-cytometry (IMC)) and profiled FFPE samples from patients with stage III colon cancer (cohort IV, $n = 29$). We previously developed and evaluated a panel of antibodies for IMC on FFPE samples for a comprehensive overview of the TME and cancer-immune cell interactions,²³ including lineage and functional immune cell markers, surrogates of cancer cell states (proliferation, apoptosis), and structural markers (epithelium, stroma, vessels). Following the data-driven identification of single-cell phenotypes, segmentation, and image analysis, we quantified the densities of major cell types. The spatial protein profiling revealed a similar cellular composition of the TME as identified using single-cell mRNA profiling (Figure S5G). Neutrophil aggregates were detectable in both, center of the tumor and the invasive margin (Figure 5G). Given the limited number of features of the IMC assay (40), the quantification of functional cell niches based on protein expression data was not possible. We therefore used a cell neighborhood analysis by considering each cell's ten nearest neighbors (STAR Methods). The analysis revealed cellular neighborhoods (Figure 5H) similar to the neighborhoods

(C) Neutrophil fractions (as percentage of leukocytes) obtained by flow cytometry ($n = 12$ patients; tumor tissue, normal-adjacent tissue, and matched blood) are shown as individual points with median $\pm$ IQR; significance was assessed using a paired t test (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

(D) UMAP of neutrophils from blood (BN) and tumor tissue (TAN) analyzed by scRNA-seq.

(E) Volcano plot of the gene expression from BNs and TANs.

(F) Expression of selected genes. UMAP colored by log normalized gene expression.

(G) Selected multiplex immunofluorescence staining of LOX-1, pan-cytokeratin, and DAPI in tumor tissue and normal-adjacent CRC tissue. LOX-1 pseudo-DAB staining and DAPI as hematoxylin in tumor and normal-adjacent colorectal cancer (CRC) tissue (right panels). Scale bars: 100 μ m.

(H) Top 10 markers for BN1-BN3 and TAN1-TAN4.

(I) Partition-based graph abstraction based on RNA velocities, projected on the UMAP plot of neutrophils from blood and tumor tissue. Edges represent cell-type transitions.

(J) scRNA-seq expression of *HLA-DRA*. UMAP colored by log normalized gene expression.

(K) Quantification of HLA-DR expression on neutrophils by flow cytometry. Each dot represents the mean value for an individual patient ($n = 12$) with indicated median $\pm$ IQR; significance was assessed using a paired t test (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

(L) Intratumoral HLA-DR⁺ neutrophils and survival in patients with early stage CRC. See also Figure S4.

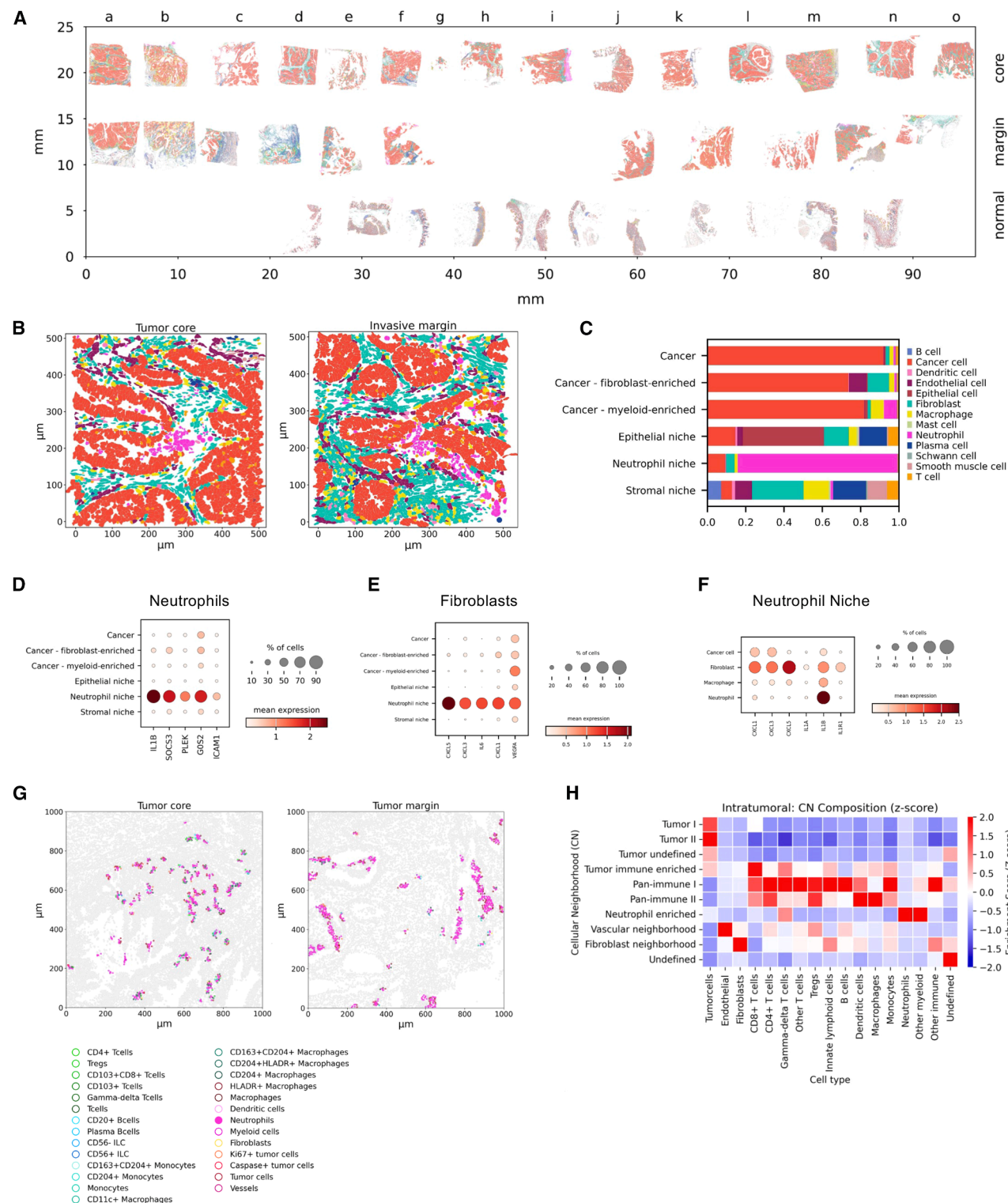


Figure 5. Spatial single-cell multi-modal profiling

(A–C) Cell types analyzed by spatial transcriptomics profiling (color key in C). (A) Overview of 37 tissue sections comprising tumor cores, invasive margins, and adjacent normal tissues from 15 patients with CRC (a–o). (B) Representative image of neutrophil clusters located in the tumor core and invasive margin. (C) Cell niches quantified using the tool NicheCompass.

(D) Top five genes upregulated in neutrophils of the neutrophil niche (compared to neutrophils within other niches; Wilcoxon test).

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reported previously.³² Importantly, our protein-based profiling confirmed the existence of neutrophil-enriched cellular neighborhoods, providing further evidence for the spatial organization of neutrophils in the TME.

Overall, the spatial single-cell transcriptomic profiling of the TME revealed cellular niches composed of different fractions of cancer cells, immune cells, endothelial cells, and fibroblasts, with distinct cell-cell communication patterns. Spatial single-cell proteomics profiling in a larger independent patient cohort confirmed the spatial organization of neutrophils in distinct niches in both the tumor core and the invasive margin. Moreover, our spatial profiling suggests a mechanism by which cancer-associated fibroblasts and neutrophils shape the tumor microenvironment and contribute to therapy resistance.

Mouse neutrophils recapitulate human neutrophil phenotypes in various compartments

Our single-cell gene expression data indicated changes in the neutrophil phenotype following migration from the bloodstream into tumors. In order to investigate the neutrophil phenotype changes along the bone marrow-blood-tumor axis, we used an orthotopic mouse model based on tumor organoid transplantation (Figure S6A). AKPS organoids – harboring the four most common CRC-driving mutations (*Apc*-LOF (loss of function), *p53*-LOF, *Kras*-GOF (gain of function), *Smad4*-LOF) – were successfully engrafted following colonoscopy-guided injection into the colonic mucosa of C57BL/6JRj mice.³³ Tumors developed within six to eight weeks and progressively expanded through the colonic layers toward the peritoneal cavity (Figures S6A–S6C), mirroring some of the invasive characteristics of late-stage human CRC. Notably, neutrophil infiltration was significantly higher in the tumor compared to the healthy colon (Figures 6A–6C). Interestingly, there was already an increase of neutrophils within the blood of AKPS-tumor bearing compared to naive mice (Figure 6C), suggesting a systemic effect during tumor development. Analysis of neutrophil activation markers such as SiglecF, CD11b, CD80, and PD-L1, further showed a progressive activation of neutrophils from the blood to the tumor (Figures 6D–6G) while MHC-II proteins were upregulated in tumor and colon tissue compared to blood neutrophils (Figure 6H). In contrast, only few phenotypic changes were observed in the bone marrow (Figures S6D and S6E). Importantly, it appears that the acquisition of Siglec-F protein expression by neutrophils occurs in the colon and at the tumor site, as there were very few Siglec-F⁺ neutrophils in the circulation and in the bone marrow (Figures 6D, S6D, and S6E). Siglec-F⁺ neutrophils are mature, long-lived tumor-promoting cells,³⁴ and the selective depletion of Siglec-F⁺ neutrophils could be of importance for the development of immunotherapies targeting myeloid cells.

Analyses of the bone marrow-tumor axis indicate tumor-induced granulopoiesis in colorectal cancer

We then performed scRNA-seq using CD45-enriched leukocytes isolated from mice six weeks post-AKPS-injection. Colons

and blood of naive (control) mice were further included as controls. Additionally, CD45-enriched leukocytes were isolated from the bone marrow of both tumor and non-tumor bearing mice. Neutrophils from all tissues and phenotypes were extracted and integrated (Figures 7A and 7B). Clustering analysis identified 11 mature and 2 immature neutrophil clusters with distinct transcriptional patterns (Figure 7C) and a high concordance of blood neutrophils and TAN marker genes derived from our human neutrophil dataset (Figure S7A). For example, *Il1a* and *Il1b* were expressed in the colon neutrophils and were at higher levels in the tumor-bearing mice compared to controls (Figures S7B and S7C). Intriguingly, *Il1b* was also expressed in the bone marrow neutrophils (Figure S7B) implicating the tumor-induced immunosuppressive programming of neutrophil progenitors as recently shown in the context of breast cancer.³⁵ Analysis of the cell cycle genes from cluster 10, a bone marrow derived cluster predominantly enriched in healthy mice, revealed that the cells are mainly in G2M phase (Figure S7D), suggesting that proliferative neutrophils reside in the bone marrow for longer periods before their recruitment to peripheral tissues without external tumorigenic signals. In the bone marrow, we could find granulocyte-monocyte progenitors (GMP) and neutrophil progenitors (ProNeutro), which already changed their transcriptional program to a more mature phenotype in tumor-bearing mice (Figure S7E). For example, GMPs expressed higher levels of the mature neutrophil markers *S100a8* and *S100a9* (Figure S7F), and ProNeutro expressed lower levels of the progenitor specific genes *Elane* and *Prtn3* (Figure S7G). We then performed pseudotime trajectory analysis, which showed a progressive transition of neutrophils from the bone marrow to the blood and colonic tissues (Figures 7D–7F). Differential gene expression analysis between tumor and healthy bone marrow and blood revealed that neutrophils already changed their transcriptional phenotype prior to their recruitment into the tumor tissue (Figures 7G and 7H).

Lastly, we carried out functional studies with the orthotopic mouse model. Following AKPS organoids transplantation, mice were treated with an anti-Ly6G antibody that specifically targets neutrophils.³⁶ This treatment did not result in overt phenotypic changes (Figure 7I). However, we observed a selective depletion of Siglec-F⁺ neutrophils within tumors, while total neutrophil numbers remained unchanged (Figure 7I). These residual neutrophils likely represent newly recruited cells from the bone marrow, consistent with previous reports.³⁶ Notably, the TME of anti-Ly6G-treated mice showed reduced numbers of exhausted CD4⁺ PD-1⁺ T cells (Figure 7G). Thus, anti-Ly6G treatment selectively depletes Siglec-F⁺ pro-tumorigenic neutrophils and promotes a more favorable TME by reducing CD4⁺ T cell exhaustion.

Together, we charted neutrophil phenotypes at the single-cell level along the bone marrow-tumor axis in a mouse model of CRC. The results suggest that neutrophils are promising targets in CRC. Further studies are necessary to investigate whether targeting IL-1 signaling to normalize tumor-induced granulopoiesis,

(E) Top five genes upregulated in fibroblasts of the neutrophil niche (compared to fibroblasts within other niches; Wilcoxon test).

(F) Selected cytokine signaling components that characterize the neutrophil niche.

(G) Neutrophil clusters identified using imaging mass-cytometry data.

(H) Heatmap of phenotypic enrichment within 10 distinct cellular neighborhoods (10 nearest neighbors, 10 neighborhoods). See also Figure S5.

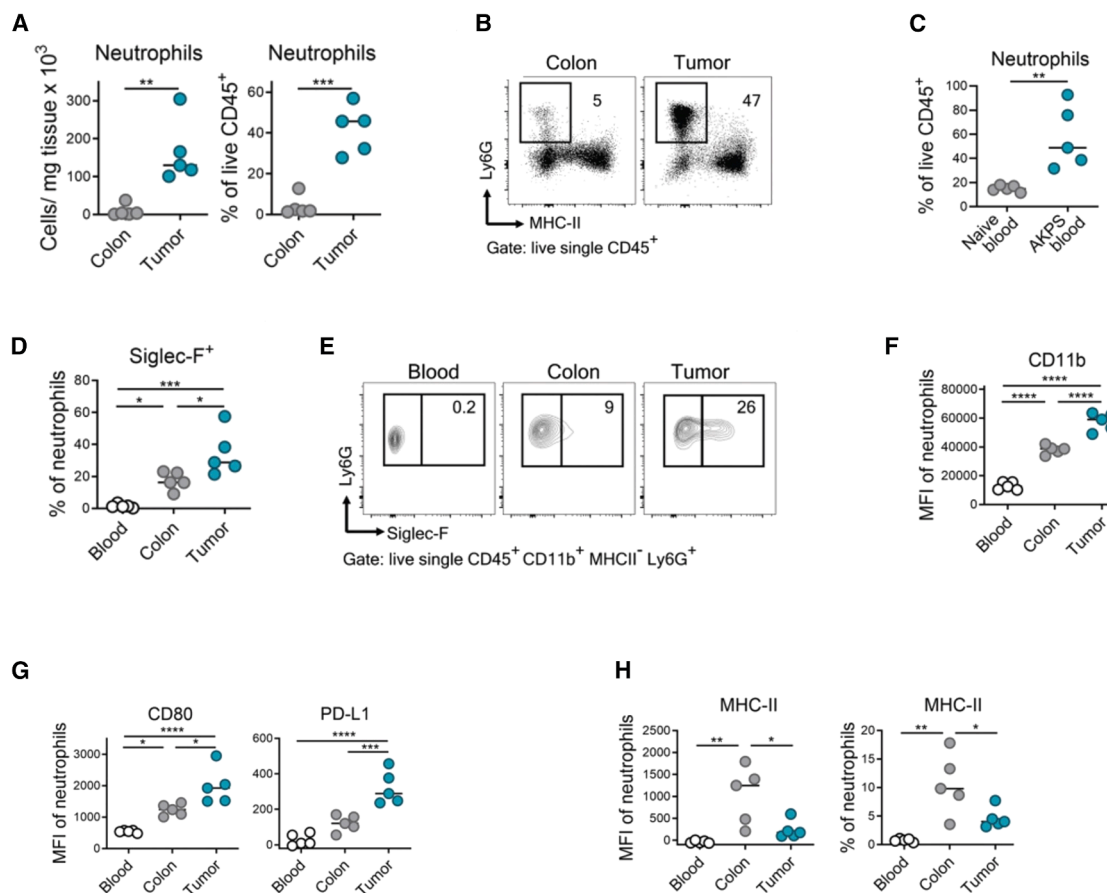


Figure 6. Neutrophils in a murine orthotopic organoid transplant model for CRC

Mouse AKPS tumor organoid cells harboring four CRC driver mutations (APC^{KO} ; $KRAS^{G12D}$; $P53^{KO}$; $SMAD4^{KO}$) were injected intra-rectally into 8- to 10-week-old C57BL/6J males.

(A) Neutrophil numbers per milligram tissue (left) and frequency among live $CD45^{+}$ leukocytes (right).

(B) Representative plots of neutrophil frequencies among $CD45^{+}$ cells, as determined by flow cytometry.

(C) Neutrophil frequencies among $CD45^{+}$ cells in the blood of naive versus AKPS-injected mice.

(D and E) Frequencies (D) and (E) representative flow cytometry plots of SiglecF-expressing neutrophils among total neutrophils in blood, colon, and tumor of AKPS-bearing mice.

(F and G) Mean fluorescence intensity (MFI) of cell-surface CD11b integrin (F) or CD80 and PD-L1 (G) expression in neutrophils.

(H) MFI (left) and frequencies of MHC II expression among neutrophils. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$; as calculated by two-tailed unpaired Student's t test for comparison of two groups (D–F) or by one-way ANOVA for statistical comparison of more than two groups (A–H). $n = 5$, medians are displayed. See also Figure S6.

or selectively depleting Siglec-F⁺ neutrophils, can be harnessed for therapy.

DISCUSSION

We built a large-scale atlas of single-cell transcriptomes of CRC through the integration of 4.27 million cells from 650 patients and 1,670 samples representing 7 billion expression values. We complemented the atlas with single-cell data from four CRC cohorts with 266 patients, and enriched the atlas with immune cells with low-mRNA content, including neutrophils, as well as with multi-modal spatial single-cell profiles of the TME comprising transcriptional profiles from 3.7 million cells, and protein profiles from 0.7 million cells. The exploitation of our unique big data resource led to several biological discoveries and might also have translational relevance.

First, we provide a high-resolution view of the TME in CRC with 61 major cell types/states and show different cell type composition patterns. The single-cell composition of the TME in CRC enabled refined tumor classification and patient stratification into four tumor immune phenotypes: immune deserted, myeloid, B cell, and T cell subtypes. Moreover, we identified consensus myeloid gene expression programs, including four immunomodulatory programs. These findings may have important implications for the development of cancer immunotherapy strategies in CRC. For example, therapies that target myeloid cells and lymphoid cells could represent an immunotherapeutic approach to treat M subtypes, as shown recently.²⁷ Similarly, given the importance of tertiary lymphoid structures,²⁸ further analysis of the B cell subtypes might open promising therapeutic avenues by targeting B cells.

Second, we characterized neutrophil subsets in various compartments in patients with CRC and showed remarkable

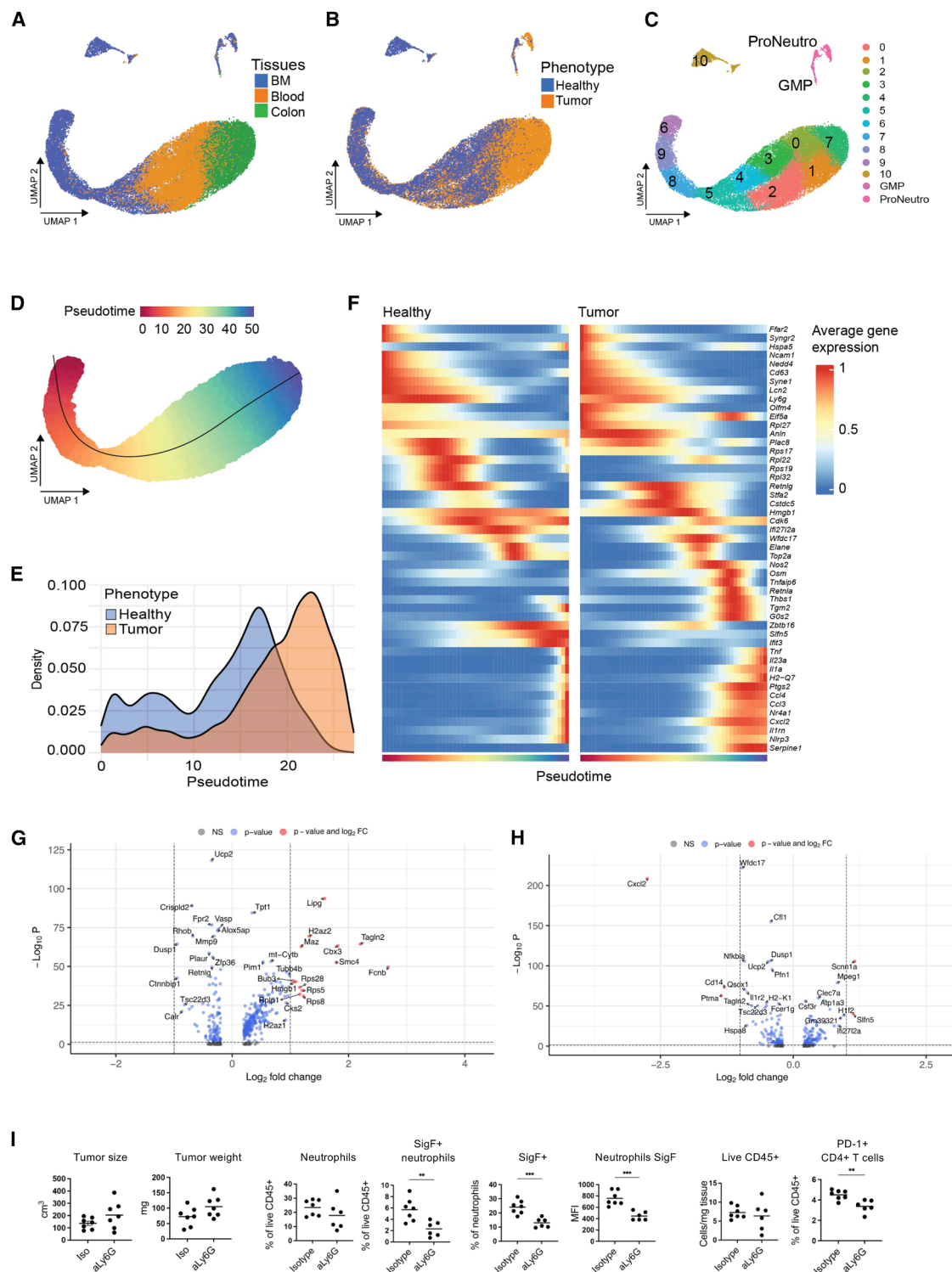


Figure 7. Neutrophil phenotypes along the bone marrow-tumor axis

(A–C) UMAP of neutrophils from tumor, colon, blood, and bone marrow of AKPS tumor bearing and non-tumor bearing mice. Cells are colored by (A) tissue type, (B) phenotype, and (C) clusters generated via the Louvain algorithm.

(D) UMAP represents pseudotime from inferred trajectory using Slingshot.

(E) Density plot comparing the pseudotemporal mapping of neutrophils between healthy and AKPS tumor-injected mice.

(F) Heatmap of genes differentially expressed between healthy and AKPS tumor injected mice (Adjusted p -values ≤ 0.05 , Wald-text, FDR correction).

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diversity and plasticity. Of particular interest for cancer immunotherapy is the HLA-DRA⁺ TAN phenotype with antigen-presenting feature that is associated with better survival outcomes. Such conversion of neutrophils to antigen-presenting cells may elicit anti-tumor immunity as recently shown in a murine model.³⁰ We show the heterogeneity of short-lived neutrophil populations in blood and long-lasting neutrophil populations in CRC tissue, which is a characteristic of a chronic inflamed microenvironment.²³ Noteworthy, using patient-derived organoids, we provide evidence for genotype-immune phenotype relationships and show that KRAS tumors polarize neutrophils into pro-tumorigenic.

Third, single-cell spatial transcriptional profiling revealed that cells build functional cellular niches with different cellular compositions. Of particular interest is a neutrophil-enriched niche in which neutrophils appear to interact with fibroblasts via IL-1 signaling. The driving force for this type of myeloid structure remains unknown and encourages further studies in the context of CRC and possibly other solid cancers.

Finally, using an orthotopic mouse model of CRC, we show that tumor-derived signals modulate neutrophils in the bone marrow, implicating tumor-induced granulopoiesis. Additionally, we also show that pro-tumorigenic Siglec-F⁺ neutrophils can be selectively depleted. Thus, harnessing the reprogramming of neutrophils for more effective therapies of cancer could be achieved by either targeting the bone marrow and/or the TME.

In conclusion, we provide a CRC atlas with single-cell resolution, which is publicly accessible and interactively explorable through cell-x-gene³⁷ (<https://crc.icbi.at>). The single-cell and spatial data we provide can be used to derive biological insights that can be subsequently validated using *in vitro* and *in vivo* models for functional studies, as we demonstrated here. For example, immune cell subtypes or cell-cell interactions between myeloid and lymphoid cells, cancer cells, and fibroblasts can be extracted, and their associations with other molecular, cellular, and clinical features can be determined. Overall, the exploitation of the high-resolution CRC atlas has the potential to provide the basis for developing combination therapies for patients with CRC who are not sufficiently responding to immune checkpoint blockers.

Limitations of the study

The integration of single-cell data across cohorts, laboratories, and platforms requires deep learning tools to remove batch effects while preserving biological variability. However, effective integration might blur subtle biological differences between related cell types and introduce unintended bias in the data. Additionally, as the majority of the previously published datasets failed to capture granulocytes in general and neutrophils specifically, the proportions of these cells in the atlas and in the respective tissues are representative only for a limited number of patients.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Zlatko Trajanoski (zlatko.trajanoski@i-med.ac.at).

Materials availability

The study did not generate any unique reagents.

Data and code availability

Data and coding scripts were deposited at <https://crc.icbi.at/> and <https://github.com/icbi-lab/crc-atlas>, respectively. Newly generated datasets were deposited at Zenodo: 16631519 (human single cell data), Biolineage Archive: S-BIAD2208 (human single cell images), and GEO: GSE285117 (mouse single cell data).

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AUTHOR CONTRIBUTIONS

Conceptualization, V.M., N.N., K.H., D.W., I.A.S. Salcher, and Z.T.; formal analysis, V.M., K.H., D.R., A.K., D.R., E.K.S., G.F., G.d.L., S.C., N.B., A.R., S.D., A.S., S. Salcher, and S. Soppor; investigation, all authors; writing – original draft, V.M., S. Salcher, and Z.T.; writing – review and editing, all authors; funding acquisition, D.W., I.A., S. Salcher, and Z.T.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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 - Patient cohorts and recruitment
 - Cell lines
 - Organoids
 - Mice
- METHOD DETAILS
 - Dissociation of normal and tumor colorectal tissue

(G and H) Volcano plots of differentially expressed genes from (G) bone-marrow and (H) blood neutrophils between healthy and tumor-bearing mice (Wilcoxon Rank-Sum test, Bonferroni corrected *p*-values).

(I) From left to right: tumor size, frequency among leukocytes, and counts per milligram (cell/mg) tissue of neutrophils, frequency of Siglec-F-positive (SigF+) neutrophils among leukocytes, and counts per milligram tissue, numbers of CD45⁺ leukocytes per milligram tissue, and frequency of PD-1⁺ CD4⁺ T cells among leukocytes. Cells were isolated from the AKPS tumors of mice treated with Isotype control or anti-Ly6G (aLy6G) antibodies. *n* = 7 Isotype, 6 aLy6G mice. Statistical test: Unpaired *t* test, means are indicated as horizontal line; *, *p* < 0.05; **, *p* < 0.01; ***, *p* < 0.001. (A–F) *n* = pool of 3 (non-tumor blood and bone marrow) or 5 (non-tumor colon and tissues from AKPS tumor-bearing mice). See also Figure S7.

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- Integration of scRNA-seq datasets
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- Patient stratification
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- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Antibodies list (human samples)	N/A	See Table S3
Antibodies list (mouse samples)	N/A	See Table S3
Biological samples		
Resected colorectal tumors and adjacent healthy colorectal tissue (CRC) samples, naive/untreated	Tirol Kliniken Innsbruck	N/A
Whole blood samples from patients undergoing colorectal tumor resections	Tirol Kliniken Innsbruck	N/A
Human neutrophils from healthy donors	Tirol Kliniken Innsbruck	N/A
Organoids from resected colorectal tumors and adjacent healthy colorectal tissue (CRC) samples	Tirol Kliniken Innsbruck	N/A
Vil-creER ^{T2} ;APC ^{KO} ;Trp53 ^{KO} ;Kras ^{G12D/WT} Smad4 ^{KO} (AKPS) organoids	Dr. Tomas Valenta and Dr. Hassan Fazilaty (DMLS, University of Zürich, Basler Lab) ³⁶	N/A
Chemicals, peptides, and recombinant proteins		
StemPro hESC SFM	Gibco	Cat# A1000701
Penicillin-Streptomycin	Sigma-Aldrich	Cat# P4333
Primocin	InvivoGen	Cat# ant-pm-2
Penicillin/Streptomycin	Sigma-Aldrich	Cat# P0781
Fetal bovine serum (FBS), sterile-filtered	Sigma-Aldrich	Cat# F7524
Dulbecco's phosphate-buffered saline, without calcium and magnesium chloride, sterile-filtered, DPBS	Sigma-Aldrich	Cat# D8537
DMEM, high glucose, pyruvate, no glutamine	Gibco	Cat# 21969-035
L-glutamine, 200 mM, 100 mL	Sigma-Aldrich	Cat# G7513
Liberase DH, research grade	Roche	Cat# 5401054001
DNAse I, grade II, from bovine pancreas	Roche	Cat# 10104159001
OptiMEM	Gibco	Cat# 31985062
Matrigel	Corning	Cat# 354277
RPMI-1640 Medium	Gibco	Cat# 11875093
RPMI-1640 Medium	Corning	Cat# 15-040-CV
HEPES	Sigma-Aldrich	Cat# H0887
Collagenase type IV	Sigma-Aldrich	Cat# C5138
Collagenase type VIII	Sigma-Aldrich	Cat# C2139
Bovine Serum Albumin, BSA	Roche	Cat# 10738328103
Penicillin/streptomycin	Sigma-Aldrich	Cat# P0781
EDTA	Sigma-Aldrich	Cat# P0781
Percoll	Cytiva	Cat# 17089101
RPMI-1640 Medium	Sigma-Aldrich	Cat# R0883
Stericup Quick Release Duapore 0.22 μm PVDF filter unit	Millipore	Cat# S2GVU05RE
anti-Ly6G	BioXCell	Cat# BE0075-1
Isotype control (IgG2a)	BioXCell	Cat# BE0089
7-AAD Staining Solution	BD Biosciences	Cat# BDB555815

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical commercial assays		
Agencourt AMPure XP magnetic beads	Beckman Coulter	Cat# A63880
BD Human Single-Cell Multiplexing Kit	BD Biosciences	Cat# 633781
BD Rhapsody Cartridge Kit	BD Biosciences	Cat# 633733
BD Rhapsody cDNA Kit	BD Biosciences	Cat# 633773
BD Rhapsody Enhanced Cartridge Reagent Kit	BD Biosciences	Cat# 664887
BD Rhapsody WTA Amplification Kit	BD Biosciences	Cat# 633801
BD Stain Buffer (FBS)	BD Biosciences	Cat# 554656
High Sensitivity D1000 Reagents	Agilent	Cat# 5067-5585
High Sensitivity D1000 Screen Tape	Agilent	Cat# 5067-5584
Qubit dsDNA HS Assay Kit	Invitrogen	Cat# Q32854
BD Pharm Lyse Lysing Buffer	BD Biosciences	Cat# 555899
Calcein AM	Invitrogen	Cat# C1430
Dracq7	BD Pharmingen	Cat# 564904
Anti-CD45 microbeads	Miltenyi Biotec	Cat# 130-052-301
BD Mouse Single-Cell Multiplexing Kit	BD Biosciences	Cat# 633793
BD Rhapsody Whole Transcriptome Analysis Amplification Kit	BD Biosciences	Cat# 633801
Qubit dsDNA HS Kit	Thermo Fisher Scientific	Cat# Q32851
TapeStation 4200 HS D5000 tapes and reagents	Agilent Technologies	Cat# 5067-5592
NovaSeq 6000 SP Reagent Kit v1.5 (100 cycles)	Illumina	Cat# 20028401
Opal 6-plex Detection Kits	Akoya Biosciences	Cat# NEL821001KT
BOND Epitope Retrieval 1	Leica Biosystems	Cat# AR9961
BOND Epitope Retrieval 2	Leica Biosystems	Cat# AR9640
BOND Dewax Solution	Leica Biosystems	Cat# AR9222
BOND Wash Solution 10x	Leica Biosystems	Cat# AR9590
Spectral DAPI	Akoya Biosciences	Cat# FP1490
Prolong Diamond Antifade	Thermo Fisher	Cat# P36961
BOND Research Detection System	Leica Biosystems	Cat# DS9455
BOND Titration Kit	Leica Biosystems	Cat# OPT9049
Xenium Human Immuno-Oncology Profiling Panel	10x Genomics	Cat# 1000654
Xenium Slides & Sample Prep Reagents	10x Genomics	Cat# 1000460
Xenium Cell Segmentation Add-on Kit	10x Genomics	Cat# 1000662
Xenium Decoding Reagents	10x Genomics	Cat# 1000461
Xenium Cassettes	10x Genomics	Cat# 1000566
Xenium Decoding Consumables	10x Genomics	Cat# 1000487
MACSxpress Whole Blood Neutrophil Isolation Kit	Miltenyi Biotec	Cat# 130-104-434
RNEasy Mini Kit	Qiagen	Cat# 74106
Luna Universal Probe One-Step RT-qPCR Kit	New England Biolabs	Cat# E3005E
T Cell TransAct human	Miltenyi Biotec	Cat# 130-111-160
Pan T Cell Isolation Kit human	Miltenyi Biotec	Cat# 130-096-535
FlexAble CoraLite Plus 555	Proteintech	Cat# KFA002

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
MUI_Innsbruck	This study	https://doi.org/10.5281/zenodo.16631519
Processed input data	This study	https://doi.org/10.5281/zenodo.16631519
Mouse single cell RNAseq data	This study	GEO accession number GSE285117
Xenium spatial transcriptomics data	This study	https://doi.org/10.6019/S-BIAD2208
Imaging Mass Cytometry (IMC) data	This study	https://doi.org/10.6019/S-BIAD2208
Bian_2018_Science (fastq)	SRA	PRJNA382695
Borras_2023_Cell_Discov (processed)	synapse	syn41687327
Burclaff_2022_Cell_Mol_Gastroenterol_Hepatol (processed)	SRA	PRJNA768275
Che_2021_Cell_Discov (fastq)	SRA	PRJNA725335
Chen_2021_Cell (processed)	synapse	syn26044148
Chen_2024_Cancer_Cell (processed)	GEO	PRJNA991601
Conde_2022_Science (fastq)	ENA	PRJEB51634
Elmentaite_2021_Nature (fastq)	ENA	PRJEB40273
GarridoTrigo_2023_Nat_Comm (fastq)	SRA	PRJNA886695
Giguelay_2022_Theranostics (processed)	SRA	PRJNA666199
Guo_2022_JCI_Insight (fastq)	SRA	PRJNA779978
Handler_2024 (processed)	GEO	PRJNA1190381
Harmon_2023_Nat_Cancer (processed)	SRA	PRJNA863562
He_2020_Genome_Biol (fastq)	SRA	PRJNA670909
HTAPP_HTAN (processed)	synapse	syn24181445
Huang_2024_Nat_Cancer (processed)	NGDC	PRJCA007578
James_2020_Nat_Immunol (fastq)	ENA	PRJEB32912
Ji_2024_Cancer_Lett (fastq)	SRA	PRJNA915211
Ji_2024_PLoS_Genet (processed)	GEO	GSE217774
Joanito_2022_Nat_Genet (fastq/processed) ³⁸	EGA	EGAD00001008555
Khaliq_2022_Genome_Biol (processed)	GEO	PRJNA827831
Kong_2023_Immunity (processed)	Broad Single Cell Portal	SCP1884
Lee_2020_Nat_Genet (fastq/processed)	GEO	PRJNA604751
Li_2017_Nat_Genet (processed)	GEO	PRJNA323703
Li_2023_Cancer_Cell (processed)	GEO	PRJNA846169
Liu_2024_Cancer_Res (processed)	GEO	PRJNA1028966
Mazzurana_2021_Cell_Res (processed)	GEO	PRJNA630996
Parikh_2019_Nature (processed)	SRA	PRJNA477814
Pelka_2021_Cell (processed)	GEO	PRJNA738517
Qi_2022_Nat_Comm (processed)	NGDC	HRA000979
Qian_2020_Cell_Res (processed)	N/A	https://blueprint.lambrechtslab.org/
Qin_2023_Cell_Rep_Med (processed)	CNGBdb	CNP0004138
Sathe_2023_Clin_Cancer_Res (processed)	N/A	https://dna-discovery.stanford.edu/research/datasets/
Scheid_2023_J_EXP_Med (processed)	Broad Single Cell Portal	SCP1690
Smillie_2019_Cell (processed)	Broad Single Cell Portal	SCP259
Terekhanova_2023_Nature (processed)	synapse	syn53214678
Thomas_2024_Nat_Med (processed)	GEO	PRJNA849977
Tian_2023_Nat_Med (processed)	Broad Single Cell Portal	SCP2079
Uhlitz_2021_EMBO_Mol_Med (processed)	GEO	PRJNA701264
deVries_2023_Nature (processed)	GEO	PRJNA894171
Wang_2020_J_Exp_Med (fastq)	SRA	PRJNA518129
Wang_2021_Adv_Sci (fastq)	SRA	PRJNA937723

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Wang_2023_Sci_Adv (processed)	GEO	PRJNA937723
Wu_2022_Cancer_Discov (processed)	N/A	http://www.cancerdiversity.asia/scCRLM
Wu_2024_Cell (processed)	N/A	http://www.pancancer.cn/neu/
Yang_2023_Front_Oncol (fastq)	SRA	PRJNA972665
Zhang_2018_Nature (processed)	GEO	PRJNA429424
Zhang_2020_Cell (processed)	figshare	https://figshare.com/articles/dataset/CRC_CD45_rda/14820318
Zheng_2022_Signal_Transduct_Target_Ther (processed)	GEO	PRJNA676119
TCGA survival data	(Liu et al.) ³⁹	N/A
TCGA data RNA-seq and masked somatic mutation data	GDC	https://portal.gdc.cancer.gov/
AC-ICAM RNA-seq data	(Roelands et al.) ²⁰	N/A
Cytosig signatures	(Jiang et al.) ⁴⁰	https://github.com/data2intelligence/CytoSig/
Experimental models: Cell lines		
HT-29	ATCC	HTB-38
LoVo	ATCC	CCL-229
HTC116	ATCC	CCL-247
CaCo2	ATCC	HTB-37
Experimental models: Organisms/strains		
Mouse: C57BL/6J Strain #:000664	The Jackson Laboratory	RRID:IMSR_JAX:000664
C57BL/6NJ mouse strain	The Jackson Laboratory	Cat# 005304 RRID: IMSR_JAX:005304
Software and algorithms		
Seven Bridges - BD Rhapsody™ WTA Analysis Pipeline	BD Biosciences	v2.0
FlowJo™	BD Biosciences	v10.7
GraphPad Prism v9	GraphPad Software	www.graphpad.com
GraphPad Prism v10.4	GraphPad Software	www.graphpad.com
Mantra Snap software version 1.0.4	Akoya Biosciences	www.akoyabio.com
inForm Tissue Analysis Software version 2.4.10	Akoya Biosciences	www.akoyabio.com
ZEN	Zeiss Software	www.zeiss.com
Cellranger	10x Genomics	v7.1.0
Xenium Onboard Analysis	10x Genomics	v3.1.0
nf-core/scrnaseq pipeline	Ewels et al., 2020 ⁴¹	v2.4.1
nf-core/maseq pipeline	Ewels et al., 2020 ⁴¹	v3.12.0
Nextflow	Di Tommaso et al., 2017 ⁴²	v23.04.0
Singularity/Apptainer	Kurtzer et al., 2017 ⁴³	v1.3.1-1.el9
Nextflow workflow and data analysis code to reproduce this study	This study	https://github.com/icbi-lab/crc-atlas
Atlas web resource	This study	https://crc.icbi.at/
Shears source code and documentation	This study	https://github.com/icbi-lab/shears
FlowJo Software	BD Life Sciences	v.10.8.0
Prism 10	GraphPad	v10
Bcl2fastq v2.20.0433	Illumina	N/A
STAR v2.5.2b	(Dobin et al.) ⁴⁴	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
R v4.3	R Core Team (2021) R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.	https://www.R-project.org/
Seurat v5.1.0	N/A	N/A
decontX v1.0.0	(Yang et al.) ⁴⁵	N/A
SingleR v2.4.1	(Aran et al.) ⁴⁶	N/A
CellDex v1.12.0	(Heng et al.) ⁴⁷	N/A
Pheatmap v1.0.12	N/A	https://cran.r-project.org/web/packages/pheatmap/index.html
Orthogene v1.8.0	N/A	N/A
Slingshot v2.10.0	(Street et al.) ⁴⁸	N/A
tradeSeq v1.16.0	(Van den Berge et al.) ⁴⁹	https://github.com/statOmics/tradeSeq
Sceasy v0.0.7	N/A	N/A
Fiji v1.54f	(Schindelin et al.) ⁵⁰	https://imagej.net/software/fiji/downloads
QuPath v0.5.1	(Bankhead et al.) ⁵¹	https://qupath.github.io/
CellProfiler v4.2.8	(Stirling et al.) ⁵²	https://cellprofiler.org/releases
R v4.4.2	R Core Team (2021) R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.	https://www.R-project.org/
Compute node CPU	Intel	Intel(R) Xeon(R) Gold 6342 CPU @ 2.80GHz (2x)
GPU node GPU	Nvidia	NVIDIA A100 80GB PCIe
Other		
Core atlas as cell-x-gene instance and h5ad/Seurat file formats with standardized metadata according to the cell-x-gene schema	This study	https://cellxgene.cziscience.com/collections/3a844375-60ea-474b-adf1-98e76928baee
pluriStrainer, 100 mm strainer, sterile	pluriSelect	Cat# 43-50100-51
CytoOne 6 cm Petri dish, uncoated	Starlab	Cat# CC7672-3359
gentleMACS Octo Dissociator with heaters	Miltenyi Biotec	Cat# 130-096-427
gentleMACS Octo Dissociator with heaters, C tubes	Miltenyi Biotec	Cat# 130-093-237
Qubit Fluorometer	Thermo Fisher Scientific	N/A
TapeStation 4200	Agilent Technologies	N/A
NovaSeq6000	Illumina	N/A
FACSymphony A5	BD Biosciences	N/A
LSRII Fortessa	BD Biosciences	N/A
Xenium Analyzer	10x Genomics	N/A
Zeiss AxioScan Z.1	Zeiss	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Patient cohorts and recruitment

Experimental data was generated using samples from four CRC patient cohorts. Patient samples from cohort I (n = 12) were used for single-cell RNA-seq. Tumor and matched adjacent normal tissues from CRC patients who underwent surgical resections at the Department of Surgery, Medical University of Innsbruck, Austria were obtained. Additionally, autologous whole blood samples were collected during surgery. Prior to study enrollment, all patients signed the informed consent for participation in the study that was approved by the ethical committee of the Medical University of Innsbruck (AN2014-0282 342/4.3 425/AM3 (4516a)). Patient samples from cohort II (n = 210) were used for mIF analysis. Samples and patient data were obtained from the University Cancer Center Frankfurt (UCT) in collaboration with the Senckenberg BioBank at the University Hospital Frankfurt. Written informed consent was obtained from all patients and the study was approved by the institutional Review Boards of the UCT and the Ethical Committee

at the University Hospital Frankfurt (project-number: SGI-11-2018). FFPE samples from cohort III (n = 15) were used for profiling using spatial transcriptomics. The study was approved by the ethical committee of the Medical University of Innsbruck (AN2014-0282 342/4.3 425/AM3 (4516a)). Patient samples from cohort IV (n = 29) were analyzed using IMC. These patients uniformly had R0-resected stage III colon cancer and received adjuvant CAPOX treatment. This analysis was approved by the Ethics Committee of the Provincial Government of Salzburg, Austria (415-E/2343/5-2018). Patients were otherwise unselected and treated at the Department of Internal Medicine III, Paracelsus Medical University, Salzburg, Austria. Human neutrophils for functional experiments were isolated from freshly collected peripheral blood of healthy donors, all of whom provided written informed consent in accordance with the study protocol approved by the Ethics Committee of the Medical University of Innsbruck (EK 1201/2024).

Cell lines

Human CRC cell lines CaCo2 (HTB-37, male), HT29 (HTB-38, female), LoVo (CCL-229, male), and HCT116 (CCL-247, male), were cultivated in RPMI 1640 medium (Corning) supplemented with 10% heat-inactivated fetal bovine serum (FBS; Sigma-Aldrich), 2 mM L-glutamine (Sigma-Aldrich), and 1% Penicillin-Streptomycin (Sigma-Aldrich) at 37°C with 5% CO₂. All cell lines were obtained from ATCC, where comprehensively authentication and quality control tests on all cell lines are performed.

Organoids

Patient-derived tumor and normal (healthy) organoids were derived from MSS CRC tumors with distinct mutational backgrounds (*KRAS* and *BRAF* mutations) and cultured in their growth media at 37°C with 5% CO₂. *Apc*-LOF, *p53*-LOF, *Kras*-GOF, *Smad4*-LOF (AKPS) tumor organoids were generated from AKP organoids and cultured in Matrigel at 37°C with 5% CO₂.

Mice

C57BL/6J (B6J, stock no. 000664, RRID: IMSR_JAX:000664) mice were obtained from The Jackson laboratory and used for injections of AKPS organoids, or kept uninjected as non-tumour bearing control mice. All mouse experiments were performed under the approval of Cantonal Veterinary Office of Zurich, Switzerland and in accordance with Swiss guidelines. All mice were maintained on a 12h light and 12h dark schedule, and water and chow were provided *ad libitum*.

METHOD DETAILS

Dissociation of normal and tumor colorectal tissue

Surgically resected human colorectal tumor and adjacent normal tissues were enzymatically and mechanically dissociated into single cells as described in detail in the following publication.⁵³ Briefly, after supervised sampling of tumor and adjacent normal CRC tissues from a pathologist, the specimens were transported to the lab on ice, washed, weighed and minced into small pieces (<1 mm). The tissue pieces were further enzymatically and mechanically digested by using DMEM medium (Gibco) with Liberase DH (Roche) and DNase I (Roche/Sigma-Aldrich) on a gentleMACS dissociator with heaters (Miltenyi Biotec) at 37°C for ~45 min (selected program: 37C_h_TDK_1). The obtained single cell solutions were sieved through a 100 µm cell strainer (pluriSelect) and kept on ice.

Generating conditioned media from patient-derived CRC organoids and cell lines

Patient-derived organoids were established as previously described.⁵³ Organoids were derived from CRC tumors with distinct mutational backgrounds and labeled according to their origin. Specifically, *KRASmut_T* refers to tumor-derived organoids harboring *KRAS* mutations and wild-type *BRAF*; *BRAFmut_T* refers to tumor-derived organoids with *BRAF* mutations and wild-type *KRAS*; and *WT_T* refers to tumor-derived organoids wild-type for both *KRAS* and *BRAF*. Corresponding normal (healthy) organoids - *KRASmut_N* and *BRAFmut_N* - were derived from independent non-tumor colonic tissue. All organoid lines used in this study were micro-satellite stable (MSS). Organoids were cultured in their respective growth media⁵³ until reaching approximately 70% confluency. CRC cell lines CaCo2 (*KRASwt*, *BRAFwt*), HT29 (*KRASwt*, *BRAFmut*), LoVo (*KRASmut*, *BRAFwt*), and HCT116 (*KRASmut*, *BRAFwt*), were cultivated in RPMI 1640 medium (Corning) supplemented with 10% heat-inactivated fetal bovine serum (FBS; Sigma-Aldrich), 2 mM L-glutamine (Sigma-Aldrich), and 1% Penicillin-Streptomycin (Sigma-Aldrich) under standard conditions until they also reached approximately 70% confluency. For conditioned medium generation, cultures were washed with phosphate-buffered saline (PBS; Sigma-Aldrich) to remove residual media and growth factors. Fresh RPMI 1640 medium, supplemented as described above, was then added, and the cells were incubated at 37°C for 48 hours to allow accumulation of secreted factors. Following incubation, supernatants were collected, centrifuged at 1,000 × g for 10 minutes at room temperature to remove debris, and filtered through a Stericup Quick Release Duapore 0.22 µm PVDF filter unit (Millipore). The filtered conditioned medium was aliquoted and stored at -20°C, with storage limited to a single freeze-thaw cycle to preserve biological activity.

Phenotypic and functional characterization of neutrophils

Human neutrophils were isolated from freshly collected peripheral blood of healthy donors using the MACSxpress® Whole Blood Neutrophil Isolation Kit (Miltenyi Biotec), according to the manufacturer's instructions. A total of 2 × 10⁶ neutrophils were cultured in 1 mL of conditioned medium. RNA expression was analyzed after 24 hours, and protein expression after 48 hours of incubation.

For quantitative RT-PCR analysis, total RNA was isolated from neutrophils cultured for 24 hours in RPMI or conditioned medium using the RNeasy Kit (Qiagen), following the manufacturer's protocol. RNA quality and concentration were assessed using a

NanoDrop 1000 Spectrophotometer (Thermo Fisher). Quantitative analysis of mRNA expression was performed using primers targeting *OLR1* (*fwd*: GAAACCCCTTGCTCGGAAGCTGA; *rev*: CAGATCCAGTCTTGCGGACAAG) and *IL1A* (*fwd*: TGTATGTGACTGCCC AAGATGAAG; *rev*: AGAGGAGGTTGGTCTCACTACC). *GAPDH* (*fwd*: GTCTCCTCTGACTTCAACAGCG; *rev*: ACCACCCTGTTG CTGTAGCCAA) served as the reference gene. Reactions were prepared using the Luna® Universal Probe One-Step RT-qPCR Kit (New England Biolabs), and amplification was performed on a QuantStudio 5 Real-Time PCR System (Applied Biosystems).

For flow cytometric analysis, neutrophils were stained with a panel of antibodies including CD3, CD14, CD16, CD45, CD181, CXCR1, and LOX-1. After staining and washing, 5 μ L 7-AAD Staining Solution (BD) was added for viability assessment. Samples were analyzed using a FACSymphony A5 flow cytometer (BD Biosciences). For T cell suppression assays, autologous neutrophils and T cells were isolated from fresh full blood of the same healthy donor. Neutrophils were prepared as described above. T cells were isolated using the Pan T Cell Isolation Kit (Miltenyi Biotec) according to the manufacturer's protocol. Neutrophils (2×10^6) were stimulated in conditioned medium for 24 hours, washed in PBS, and then co-cultured with T cells at a neutrophil to T cell ratio of 3:1. Upon start of co-culture, T cells were activated concurrently using T Cell TransAct (Miltenyi Biotec) at 5 μ L/mL concentration. After 24 hours of co-culture, expression of the early activation marker CD69 on CD4⁺ and CD8⁺ T cells was assessed by flow cytometry using a defined antibody panel (CD3, CD4, CD8, CD16, CD69). 7-AAD Staining Solution was added for viability assessment, and samples were analyzed on a FACSymphony A5 flow cytometer (BD Biosciences). All antibodies and reagents used for flow cytometry are listed in the [key resources table](#). Details on antibodies are provided in [Table S3](#) related to STAR Methods Data were analyzed using FlowJo software version 10.7 (BD Biosciences).

BD Rhapsody library preparation and sequencing

Freshly isolated single-cells were immediately processed with the BD Rhapsody scRNA-seq platform (BD Biosciences). Red blood cells were removed from dissociated tissues and from whole blood samples using the BD Pharm Lyse lysing solution (BD Biosciences). Cell number and viability were assessed with the BD Rhapsody scRNA-seq platform using Calcein-AM (Invitrogen) and Draq7 (BD Biosciences). The BD Single-Cell Multiplexing Kit (BD Biosciences) was used to combine and load multiple samples (human colorectal tumor tissue, adjacent normal tissue, blood) onto a single BD Rhapsody™ cartridge (BD Biosciences). Sample-tag staining was performed according to the manufacturer's protocol (sample-tag staining at room temperature for 20 min and washing by centrifugation at 400 g for 5 min). Single-cell isolation in microwells (cell load: 20 min incubation at room temperature, 20°C $\pm$ 2°C) with subsequent cell-lysis and capturing of poly-adenylated mRNA molecules with barcoded, magnetic capture-beads was performed according to the manufacturer's instructions. Beads were magnetically retrieved from the microwells, pooled into a single tube before reverse transcription. Unique molecular identifiers (UMIs) were added to the cDNA molecules during cDNA synthesis. Whole transcriptome amplification (WTA) and sample-tag sequencing libraries were generated according to the BD Rhapsody single-cell whole-transcriptome amplification workflow. The quantity and quality of the sequencing libraries was analyzed with the Qubit dsDNA HS (High Sensitivity) assay kit (Invitrogen) and the 4200 TapeStation (Agilent) system. Libraries were sequenced on the Novaseq 6000 system (Illumina) targeting a sequencing depth of 50,000 reads/cell.

Generation of the atlas

The single-cell atlas was compiled from publicly available datasets,^{17,54–99} including data from previously published colorectal cancer (CRC) studies, our own data (cohort I), and data from 12 additional non-CRC studies for control purposes, including only healthy, non inflamed control colon samples from adult individuals (see [Table S1](#)). These studies, published between May 2017 and July 2024, utilized six different sequencing platforms, including the widely used 10x Chromium (10x Genomics) as well as Smart-seq2,¹⁰⁰ SMARTer (C1), scTrio-seq2, GEXSCOPE (Singleron), TruDrop¹⁰¹ and DNBelab C4.¹⁰² Our own data was generated using the micro-well-based BD Rhapsody scRNA-seq platform (BD Biosciences). We carefully chose studies that followed comparable protocols for sample processing and data generation, such as sequencing of whole cells, and had raw gene expression data as well as cell-level metadata available to map cells to sample information. Studies using flow cytometry-based cell sorting prior to sequencing were not excluded, as they offer important information about rare cell types.^{103–105} To distinguish between normal colon samples from CRC patients and healthy individuals, we use the terms “healthy normal” for samples from healthy individuals and “adjacent normal” for colon samples near tumor tissue.

Preprocessing and quality control of scRNA-seq data

We distinguish between studies, which are scientific publications, and datasets, which consist of scRNA-seq samples processed with the same sample preparation and experimental platform. A study can include one or more datasets. Fastq files of our cohort I dataset were processed using the Seven Bridges - BD Rhapsody™ WTA Analysis Pipeline version 2.0. In addition, fastq files were available for samples from the studies Bian_2018_Science, Che_2021_Cell_Discov, Conde_2022_Science, Elmentaite_2021_Nature, GarridoTrigo_2023_Nat_Comm, Guo_2022_JCI_Insight, He_2020_Genome_Biol, James_2020_Nat_Immunol, Ji_2024_Cancer_Lett, Joanito_2022_Nat_Genet, Lee_2020_Nat_Genet, Wang_2020_J_Exp_Med, Wang_2021_Adv_Sci, Yang_2023_Front_Oncol and were obtained from the identifiers specified in the [key resources table](#). Smart-seq2 data was processed using the nf-core RNA-seq pipeline v3.12.0 (<https://doi.org/10.5281/zenodo.7998767>), 10x datasets were processed with cellranger v7.1.0 (10x Genomics) using the nf-core scRNA-seq pipeline v2.4.1⁴¹ (<https://doi.org/10.5281/zenodo.8399014>) with the GRCh38 reference genome and GENCODE v44 annotations. All other datasets were obtained as count tables from their respective identifiers. All raw count matrices were loaded into AnnData¹⁰⁶ and further processed with scverse tools.¹⁰⁷ For

samples with available fastq files, ambient RNA was removed using scAR¹⁰⁸ as implemented in scvi-tools.¹⁰⁹ Quality control was performed using scanpy¹¹⁰ by retaining cells with (1) >400 transcripts, (2) >100 genes, and (3) <50 % mitochondrial transcripts, followed by 3 MADs (median absolute deviations) filtering on individual samples.¹¹¹

Integration of scRNA-seq datasets

Individual datasets were merged into a single AnnData object. Since genome annotations partly differed between the datasets, we mapped gene symbols to the ensembl ids of the used annotation version and merged on stable ensembl ids. If gene ids were missing from a dataset, the values were filled with zeros. Genes that were missing in more than 19 datasets (25%) were excluded altogether.

We integrated the datasets using the scANVI algorithm,^{109,112} as it has been demonstrated to be one of the top-performing methods for atlas-level integration and to scale to >1M cells.¹¹³ Given that scANVI requires cell-type annotations for one input dataset, we manually annotated the major sample types, including tumor and polyp, normal, metastasis, blood, and lymph nodes, as these were expected to represent the main sources of biological variation. These five “seed” datasets were annotated based on unsupervised clustering as described below. Raw counts were used as input for scANVI. Counts from plate-based assays such as Smart-seq2 were scaled by the gene length as recommended on the scvi-tools website. The scANVI model was initialized with a pre-trained scVI model,¹¹⁴ as recommended in the scvi-tools tutorial. The scVI model was trained on the 6000 most highly variable genes as determined with scanpy’s¹¹⁰ *pp.highly_variable_genes* with parameters *flavor*=“seurat_v3” and *batch_key*=“dataset”. Each sample was considered as an individual batch for both scVI and scANVI. Other than that the algorithms were run with default parameters.

Unsupervised clustering and cell-type annotation

We computed UMAP embeddings¹¹⁵ and unsupervised Leiden-clustering¹¹⁶ with scanpy,¹¹⁰ based on a cell-cell neighborhood graph derived from scANVI latent space. Coarse, lineage-specific clusters were iteratively sub-clustered to identify cell-types at a more fine-grained resolution. Cell type clusters were annotated based on previously reported marker genes.⁶⁰

Patient stratification

We stratified patients into immune phenotypes based on immune cell-type fractions. We selected all patients with primary tumor samples and excluded enriched datasets. Neutrophil fractions were excluded, since they are not appropriately captured in the majority of datasets. Cell-type fractions of primary tumor samples were loaded into a patient × cell-type AnnData container. Dataset-specific batch-effects were removed using a linear model as implemented in *scanpy.pp.regress_out*. Patients were clustered using graph-based Leiden clustering with the “correlation” distance metric for computing the neighborhood graph. Patient clusters were labeled according to their predominant cell-types.

Consensus myeloid expression programs

We applied non-negative matrix factorization (NMF),¹¹⁷ following Miller et al.,¹⁸ to identify identity and activity gene programs in myeloid cells. After filtering for tumor, untreated samples, ≥20 cells per myeloid subtype, and genes expressed in ≥200 cells with scanpy,¹¹¹ we retained 100,082 cells from 14 datasets. cNMF was run cohort-wise on batch-corrected counts (*Preprocess.preprocess_for_cnmf()* function,¹¹⁷ testing K = 7–19 with 150 iterations. Optimal K was selected per dataset based on silhouette score and error. Consensus programs were derived by clustering gene spectra cosine similarity¹¹⁸ using Ward’s method,¹¹⁹ merging programs with cophenetic distance ≤3. We obtained 17 consensus programs, annotated using *decoupler*¹²⁰ via ORA¹²¹ on the top 20 genes with MSigDB^{122,123} GO terms^{124,125} and via GSEA¹²² with Hallmark gene set,¹²³ considering also program usage across datasets and ‘cell_type_fine’ labels.

Differential gene expression testing

We used DESeq2¹²⁶ on pseudo-bulk samples for differential expression testing which has been demonstrated to perform well and properly correct for false discoveries.¹²⁷ For each cell type and sample, we summed up transcript counts for each gene that is expressed in at least 5% of cells using *decoupler-py*.¹²⁰ Pseudo-bulk samples consisting of fewer than 10 cells were discarded. We compared primary tumor samples from tumor vs. normal, tumor vs. blood (*sample type*), primary tumor samples from the patient subtypes M/B/T/desert (*group*), BN vs. TANs (*cell_type_fine*), and Neutrophil clusters (BN1-3, TAN1-4), including the patient as a covariate for paired comparisons, else we included the dataset. For comparisons between multiple groups, we used contrasts with sum-to-zero coding. P-values were adjusted for multiple hypothesis testing with independent hypothesis weighting (IHW).¹²⁸

Liana+ of scRNA-seq data

We performed liana analysis on the atlas using LIANA+²⁸ combining DEA results from HLA-DR+ neutrophils. Ligand–receptor interactions were inferred using likelihood ratios (consensus, threshold 0.1) For visualization, we generated a dot plot summarizing the ligand–receptor interactions. *ligand_pvalue* reflects ligand significance in DEA, while *Interaction_stat* integrates expression levels and statistical significance.

RNA velocity analysis

We performed RNA-velocity analysis on our own dataset (cohort I) using velocity and scVelo. BAM files generated by the BD Rhapsody WTA analysis pipeline were preprocessed with samtools to ensure compatibility with velocity (see preprocessing/bd_rhapsody/velocity.nf in our git repository for more details). The resulting loom files from velocity were then processed in scVelo to estimate and visualize RNA velocities, following the scVelo tutorial. To reconstruct higher-level transitions between cell types, we employed Partition-based Graph Abstraction (PAGA) using the RNA velocity-derived graph. The PAGA graph was computed with `minimum_spanning_tree=False`, allowing for all high-confidence connections to be retained without forcing a tree structure. The use of RNA velocity as the underlying connectivity graph (rather than transcriptomic similarity alone) integrates dynamic information into the inferred trajectories. We grouped cells by manually curated cell subtypes (`cell_type_coarse`) and visualized the PAGA transitions in the original UMAP space. Because of a known issue in the current version of scVelo, we manually ensured consistency between `obs["distances"]` and `uns["neighbors"]`, which PAGA internally relies on. Transition confidences between cell types were extracted using `scv.get_df(adata_merged, "paga/transitions_confidence")`, and no thresholding was applied to ensure a comprehensive visualization of all potential paths (`threshold=0`).

Mapping bulk RNA-seq data in the atlas using shears

We developed Shears, a Python-based method designed to address critical limitations of existing algorithms^{129,130} to associate single cells from scRNA-seq datasets with phenotypic or clinical information of bulk RNA-seq samples (source code and documentation are available at <https://github.com/icbi-lab/shears>). For example the widely used developed method Scissor¹²⁹ has several limitations: 1) The method is computationally expensive, as demonstrated by the several thousand CPU hours required to apply it to large single-cell datasets like the CRC atlas; 2) It does not adequately account for biological replicates within the single-cell data; and 3) Despite being based on a generalized linear model it is not possible to include covariates into the regression analysis. Hence, it is not possible to determine whether the observed effects are due to confounding factors. For example, when analyzing the entire dataset for associations with a mutation that occurs preferentially in COAD over READ, it is unclear whether the 'scissor cells' are linked to the mutation itself or the tumor type. This uncertainty extends to other variables as well, such as gender, age, microsatellite instability, or co-mutations. Additionally, Scissor results are determined relative to other cells in the single-cell RNA-seq dataset. The cells that show the strongest association with the phenotype of interest are classified as Scissor-positive or Scissor-negative while L1-regularization forces other coefficients of less important cells to be zero. As a consequence, in large datasets, Scissor results show the average of which cells were chosen as "most important" over many patients. To overcome these limitations, we developed Shears, an alternative method that enhances computational efficiency for large datasets (>1M cells) while addressing the handling of covariates and biological replicates, enabling more accurate identification of cell subpopulations from single-cell data associated with a given phenotype.

Shears utilizes a two-step approach: 1) generate a bulk x single cell matrix with cell weights assuming that $\text{bulk} = \text{weights} * \text{single cells}$. This is similar to deconvolution, except we have a single-cell matrix instead of a signature matrix aggregated by cell types and can be solved with e.g. linear regression. To ensure comparability between bulk and single-cell data, quantile normalization is applied, as done by Scissor. Given the large number of features, we apply a Ridge constraint to the regression. This Ridge regression estimates weights for the following equation: $B = w_1 * S_1 + w_2 * S_2 + \dots + w_n * S_n$. In contrast to the correlation matrix used by Scissor, by utilizing a weight matrix, we can account for the relative importance of each cell, thereby reducing the risk of falsely attributing an association between a cell and the phenotype when its effect could actually be explained by other cells. 2) Compute the importance of each cell for a given phenotype using an individual linear model per cell and fitting a model with "Phenotype ~ cell weight + covariates" for each cell. Shears uses logistic regression for mutation data and Cox regression for overall survival. The coefficient of the model can be considered as the effect size measured for each cell, while the p value provides evidence on whether the effect size for each cell is statistically significant in explaining the phenotype. Shears accounts for biological replicates by grouping data by biological replicate (e.g., patient or sample) and cell type, ensuring at least 10 cells per group and cell type, then calculating the mean coefficient weighted by $-\log_{10}(\text{p-value})$ to reflect both effect size and statistical significance of the phenotype on a given cell. Significant differences are determined by comparing the fractions of shears+ and shears- cells with a paired wilcoxon test with `zero_method = "zsplit"` as implemented in the scipy package. p-values were Benjamini-Hochberg-adjusted and considered significant at an FDR <0.01.

Shears was used to associate phenotypic data from the combined TCGA and AC-ICAM bulk RNA-seq cohorts with our single-cell data. TCGA COAD and READ cohort masked somatic mutation data and gene expression data was obtained from the GDC portal, survival data from Liu et al., 2018.³⁹ AC-ICAM mutation, survival, and gene expression data was obtained from the paper supplements. For mutations Shears was run with the covariates dataset (AC-ICAM/TCGA) and type (COAD/READ). Shears for survival data was run with the covariates tumor_stage (early/late), age_scaled, and dataset.

Flow cytometry staining of human samples

Cells isolated from surgically resected CRC tumor and adjacent normal tissues as well as whole blood obtained just prior surgery were stained with a cocktail of the following 20 antibodies (CD3, CD8, CD4, CD14, CD16, CD19, CD15, CD28, CD34, CD31, CD38, CD45, CD56, CD90, CD123, CD161, CD193, CD326, HLA-DR, TCRgd) to define all cell populations of interest. After washing and addition of 5 μL 7-AAD, cells were measured on a FACSymphony A5 flow cytometer (BD Biosciences). All source data of the

aforementioned antibodies is provided in the [key resources table](#), applied fluorochromes are listed in [Table S3](#). Data were analyzed using FlowJo v10.7 software. For details of the gating strategy, see [Figure S8](#).

Multiplex immunofluorescence of human samples

CRC tumor and tumor-adjacent tissue samples were fixed in 4% formalin and embedded in paraffin. Three-micrometer sections were used for immunofluorescence staining. The immunofluorescence staining on formalin-fixed paraffin-embedded (FFPE) tissue was conducted using the Opal 6-Plex Detection Kit (Akoya Biosciences). A multiplex panel of immune markers was established utilizing antibodies against CD15 (1:200, clone MMA, Abcam), LOX-1/OLR1 (1:200, polyclonal, Sigma-Aldrich), and cytokeratin (1:1000, clone AE1/AE3, Dako; clone C-11, Abcam). The staining procedure was performed using an automated staining system (BOND-RX, Leica Biosystems). Markers were applied sequentially and paired with their respective Opal fluorophores (CD15 – Opal650, LOX-1/OLR1 – Opal570, cytokeratin – Opal690). To visualize cell nuclei, the tissue was stained with 4',6-diamidino-2-phenylindole (spectral DAPI, Akoya Biosciences). The stained slides were scanned using the Mantra 2 Quantitative Pathology Workstation (Akoya Biosciences), and representative images were acquired using the Mantra Snap software version 1.0.4. Spectral unmixing, multispectral image analysis, and cell phenotyping were carried out using the InForm Tissue Analysis Software version 2.4.10 (Akoya Biosciences). Pseudo-DAB staining projections were achieved by converting fluorescent channels to pseudo-brightfield images using InForm, where fluorescent markers are displayed as pseudo-DAB staining and DAPI as hematoxylin.

Sample preparation for xenium *in situ* workflow

Sections were prepared according to manufacturer's instructions (Tissue Preparation Guide Demonstrated Protocol CG000578) at the Department of Anatomy, Histology and Embryology, Medical University of Innsbruck. 5 μ m sections were sectioned from FFPE tissue blocks with a microtome (Eprelia, HM355S), floated in a 45°C water bath, and adhered to Xenium slides (10x Genomics). Samples were baked at 42°C for 3 hours and stored in a desiccator at room temperature (20°C $\pm$ 2°C), overnight, followed by deparaffinization and decrosslinking, according to manufacturer's instructions (Deparaffinization & Decrosslinking Demonstrated Protocol CG000580).

Xenium *in situ* workflow and data acquisition

Xenium samples were processed according to manufacturer's protocols (Xenium *In Situ* Gene Expression with Morphology-based Cell Segmentation Staining User Guide CG0000749 and Decoding & Imaging, User Guide CG000584). Samples were stained utilizing the predesigned Human Immuno-Oncology panel from 10x genomics (PN 1000654), as they became available from the manufacturer. Slides were stained with a Xenium Cell Segmentation kit according to manufacturer's instructions (User Guide CG0000749). Briefly, probes of the immuno-oncology panel were hybridized overnight before rolling circle amplification and native protein autofluorescence was reduced with a chemical autofluorescence quencher. Slides were processed on a 10x Xenium Analyzer, with ROIs selected to cover the selected regions. The Xenium Onboard Analysis pipeline v2.0.0 (10x Genomics) was run directly on the instrument for imaging processing, cell segmentation, image registration, decoding, deduplication, and secondary analysis, as previously described.¹³¹

H&E staining of human samples

After processing tissue sections through the Xenium workflow, the same sections on the Xenium slide were used for H&E staining, according to the manufacturer's instructions (Post-Xenium Analyzer H&E Staining CG000613) and imaged using a PANNORAMIC slide scanners (Institute for Pathology, Neuropathology & Molecular Pathology, Medical University of Innsbruck).

Immunohistochemistry of human samples

Consecutive FFPE tissue slices of CRC tumor core, invasive margin and adjacent normal samples (n = 11) were stained with the following antibody for immunohistochemistry (IHC): CONFIRM anti-CD15 (clone MMA, Mouse Monoclonal, Primary Antibody, ready-to-use (RUI), IVD (Ventana, Roche). Staining was made on a BenchMark Ultra instrument (Ventana, Roche). The whole slides were scanned with the VENTANA® DP 200 slide scanner (Ventana Roche) and the navify® Digital Pathology platform to retrieve digital slide tissue images. Digital slides were further processed with the QuPath software v0.5.1 and ImageJ software v1.54g.

Post Xenium and H&E staining, CRC tumor core and invasive margin samples (n = 5) were stained with CD15 antibody and assessed by IHC on an automated staining device (DAKO Omnis, Agilent) by using a ready-to-use kit (mouse anti-human, clone Carb-3, Agilent). Normal human tonsil and normal skeletal muscle were included as positive and negative controls in each staining run.

Xenium *in situ* data analysis

Sample areas were manually selected on the Xenium Analyzer or in the Xenium Explorer software. Xenium output bundles were further processed with scanpy,¹¹⁰ squidpy¹³² and the SpatialData framework.¹³³ Samples from different slides were merged and spatially aligned. Cells were filtered to have a minimum number of 15 transcripts, less than two control codewords, and at least four spatial neighbors within 50 μ m, resulting in a dataset of 3.7 million cells. Integration of samples was done on the filtered counts matrix with an scvi model of one hidden layer and 30 latent components. Leiden clustering was performed on the integrated latent space. To annotate leiden clusters, we used a three-step approach: First, we utilized tangram¹³⁴ in cluster mode to predict cell type labels with both a downsampled

version of the complete CRC atlas and with only the MUI cohort, which consists of single-cell RNA sequencing samples from matched patients. scSampler¹³⁵ was used for downsampling of the complete atlas. To generate a consensus annotation, we mapped the middle and fine cluster annotations from the references to single cells of the spatial atlas, using the batch-corrected counts as expression profiles. Leiden clusters at resolution 2.5 were then annotated by the most abundant predicted middle cell type label and at resolution 3.8 by the most abundant predicted fine cell type label. Second, we used a Wilcoxon test to calculate the top markers per cluster to identify cell types not present in the reference data (i.e., smooth muscle cells). Third, we checked the consistency of the annotated cell types with their spatial localization. To identify cell niches, NicheCompass¹³⁶ was run on the raw counts with the 'gcnconv' encoder and each sample as batch factor. OmniPath¹³⁷ and NicheNet¹³⁸ were used as prior gene programs. The input spatial neighbors graph was calculated with squidpy using Delaunay triangulation for cell centroids within a radius of 50 μm . Leiden clustering was then performed at different resolutions on the niches latent space to identify distinct spatial niches. Marker genes for each cell type within a given niche were obtained using a Wilcoxon test with pre-test filtering to remove spillover transcripts from adjacent cells. Neutrophil clusters were identified from the spatial neighbors graph by first defining cluster seeds as cells of cell type 'Neutrophil' with at least three homotypic neighbors. Cluster seeds were subsequently expanded in ten iterations to add all neutrophils adjacent to seed cells to the respective cluster. To investigate cell-cell-communication pathways within cell niches, the steady-state version of LIANA+²⁸ was run for cell types with at least 100 cells in a given niche, consensus resources, default methods and 100,000 permutations.

Tissue microarrays and IF image acquisition

Five tissue microarrays (TMAs) containing a total of 549 valid cores were processed and stained for HLA-DR and CD15 using immunofluorescence (IF). In brief, slides have been deparaffinized, processed by heat-induced epitope retrieval and incubated with anti-HLA-DR (1:250, Abcam) and anti-CD15 (Dako/Agilent). Staining has been visualized by fluorochrome-labeled secondary antibodies (ThermoFisher); for HLA-DR staining, HLA-DR antibody has been pre-labeled using FlexAble CoraLite Plus 555 (Proteintech) according to the manufacturer's instructions. Images of the stained TMAs and whole slides were acquired in .czi format using AxioScan Z.1 (Zeiss).

IF image preprocessing and core extraction

The .czi images were converted to a compatible format (.tif) for downstream analysis using Fiji (ImageJ, v1.54f). Images were carefully inspected during this step to ensure no loss of resolution in any of the RGB layers. Individual TMA cores were extracted from the pre-processed images using QuPath (v0.5.1), an open-source software platform for digital pathology. The cores were manually annotated to exclude damaged or improperly stained regions, and only valid cores were included in subsequent analyses.

Cell identification and quantification in IF images

A customized image analysis pipeline was developed in CellProfiler (v4.2.8) to identify and quantify nuclei and marker-positive cells for CD15 and HLA-DR. The pipeline consisted of the following steps: 1. Nuclear Identification: Nuclei were segmented and identified based on fluorescence intensity.

2 Marker Co-Localization: Positive staining for CD15 and HLA-DR was identified and quantified, with nuclei serving as reference points for spatial localization and co-expression analysis.

Output: The pipeline produced quantitative data on nuclei counts, their spatial position and the number of nuclei positive for each marker, based on staining intensity to distinguish specific marker expression from background fluorescence. Quantitative data, including total nuclei count, all CD15-positive cells, all HLA-DR-positive cells and co-expression counts, were exported from CellProfiler for statistical analysis.

Imaging mass cytometry and data analysis

Imaging mass cytometry was performed on tissue microarrays from microsatellite-stable CRC patients as previously described,²³ using the antibody panel provided in Table S3. Images were obtained from a total of 37 patient samples, including intratumoral regions from 27 patients and tumor margin regions from 29 patients. Among these, both intratumoral and tumor margin regions were imaged for 19 patients. Per sample, one or two 1000x1000 μm regions of interest were ablated, depending on tissue availability. Images were visually inspected and exported to ome.tif using MCD viewer (standard biotools). Image normalization, cell segmentation and phenotype identification were performed as described previously²³ and steps were validated throughout on the original images.

Spatial analysis of neutrophil niches for IF and IMC data

To analyze the spatial organization in tumor and margin regions, we constructed Voronoi diagrams based on the center coordinates of all detected nuclei. Each Voronoi region was then assigned the corresponding cell type. The DBscan clustering algorithm was used to find niches consisting of at least 3 neutrophils (granulocytes) in a maximum pairwise distance of 15 μm . For these niches all other cell types that share a Voronoi ridge (edge) with a neutrophil were searched and added as "niche members". To determine if the number of detected niches differed significantly from what would be expected by chance, a Monte Carlo simulation with 1,000 iterations was conducted. In this simulation, the cell type labels for each cell on the imaged slide was randomly assigned while maintaining the number of cells for each type constant and preserving the overall spatial arrangement. A z-score and p-value were then calculated (see Table S4).

Cellular neighborhood analysis of IMC data

Cellular neighborhoods (CNs) were defined using spatial analysis of IMC data following the approach originally described in Schürch et al.²⁹ For each cell, a localized “window” was generated by selecting its 10 nearest cellular neighbors (including the cell itself) based on Euclidean distance within its corresponding IMC image. These compositional windows were then converted into vectors representing the frequency of each constituent cell phenotype. These phenotypic composition vectors were clustered in an unsupervised manner using the MiniBatchKMeans algorithm (Scikit-learn v1.5.2) with $k=10$ clusters (random state 42), thereby assigning each cell to a distinct CN. Phenotypic enrichment within these defined cellular neighborhoods was visualized in a heatmap using the z-score derived from fold-change values. These fold-change values represent the ratio of a given cell type’s proportion within a specific CN to its overall proportion across the entire tissue. The z-score, calculated across the 10 CNs for each cell type, thus indicates the degree to which a cell type is statistically over- or under-represented in a particular CN relative to its distribution in other CNs.

Survival analysis

Survival analysis was performed to assess the impact of the number of neutrophil niches on overall survival. Patients were stratified into “high” and “low” groups based on an optimal cutoff that minimizes the p-value or by or by setting it to 0.66. Kaplan-Meier survival curves were generated for each group using the KaplanMeierFitter function from the lifelines Python package. The log-rank test, implemented via the logrank_test function from the same package, was used to compare survival distributions between the two groups. Cox proportional hazards regression was employed using the CoxPHFitter function (lifelines) to calculate hazard ratios (HR) and 95% confidence intervals (CI).

Colonoscopy submucosal injection of AKPS organoids into mice

Apc-LOF, *p53*-LOF, *Kras*-GOF, *Smad4*-LOF (AKPS) tumor organoids were generated from AKP organoids harbouring a RNP-mediated *Smad4* mutation introduced by CRISPR-Cas9.³³ Organoids were cultured in Matrigel and passaged every 2–3 days.¹³⁹ Colonoscopy-guided orthotopic cell transplantation was done as described previously with modifications.¹⁴⁰ Briefly, following mechanical dissociation, the equivalent of one Matrigel done was resuspended in 50 mL OptiMEM (Gibco) to equate to one injection in one mouse. 8- to 10-week-old males were anesthetized with isoflurane and placed facing up on a 37°C heating pad. Fecal content was flushed from the colon with warm DPBS and AKPS organoid solution was injected carefully below the mucosa using a custom needle, syringe and colonoscope. Observation of a visible injection bubble was noted as an indicator for successful injection in the correct location (Figure S6A). Mice were then regularly monitored according to official guidelines until experimental endpoint.

Tissue dissociation of mouse samples

AKPS tumors: Tumors and disseminated tumors from the peritoneal cavity of 5 mice were collected at the study endpoint, pooled respectively and cut into small pieces. Tissue was digested at 37°C for 50 min in RPMI-1640 containing 15 mM HEPES (H0887 Sigma), 500 U/mL^{−1} type IV (C5138 Sigma) and type VIII collagenase, (C2139 Sigma) and 0.05 mg/mL DNase I (10104159001 Roche). Non-digested tissue was mechanically disrupted with a syringe and cell suspension was pushed through a 70 μm cell strainer. Flow-through was topped with DPBS buffer containing 2% BSA, 100 U mL^{−1} penicillin/streptomycin (P0781 Sigma) and 5 mM EDTA and cells were spun prior and resuspended prior analysis.

Blood from naïve and AKPS tumor-bearing mice: Blood was collected from mice post-mortem by cardiac puncture in 2% BSA 5 mM EDTA, followed by red blood cell lysis with distilled cold water for 30 sec. Blood from 3 naïve and 5 AKPS tumor-bearing mice were pooled respectively.

Colon from naïve mice and adjacent colon to AKPS tumor: Colonic tissue from 5 mice per condition was collected, cleaned and opened longitudinally, prior to washing with PBS. Tissues were cut into small pieces, pooled per condition and washed twice while shaking for 25 min at 37°C with HBSS 2% BSA, 100 U mL^{−1} penicillin/streptomycin (P0781 Sigma) and 5 mM EDTA. Tissue was digested at 37°C for 50 min in RPMI-1640 containing 15 mM HEPES (H0887 Sigma), 500 U/mL^{−1} type IV (C5138 Sigma) and type VIII collagenase, (C2139 Sigma) and 0.05 mg/mL DNase I (10104159001 Roche). Non-digested tissue was mechanically disrupted with a syringe and cell suspension was pushed through a 70 μm cell strainer. Cells were centrifuged for 8 min and layered onto a 40/80% Percoll (17089101 Cytiva) gradient (18 min, 2.100 g, 20°C, no brake). The interphase was collected and washed in DPBS.

Bone marrow from naïve and AKPS tumor-bearing mice: Tibia and femur were collected and flushed with RPMI medium using a 23-gauge needle. The content was filtered through a 40 μm strainer followed by red blood cell lysis via incubation in ice-cold distilled water for 30 sec. Bone marrow from 3 naïve and 5 AKPS tumor-bearing mice were pooled respectively.

Antibody neutralization

Three weeks after orthotopic AKPS CRC organoids injection, tumor-bearing mice (male B6NRJ) were injected intraperitoneally twice a week with 0.5 mg anti-Ly6G (BE0075-1 BioXCell, 1A8) or anti-trinitrophenol isotype control (BE0089, BioXCell, 2A3).

Sample processing for flow cytometry

Tumors were collected at experimental endpoints and were minced and digested in a shaking incubator at 37°C. The digestion mix consisted of RPMI 1640 supplemented with 10% FBS, 100 U/ml penicillin/streptomycin (P0781, Sigma), an equal amount of 250 U/ml type IV (C5138 Sigma) and type VIII collagenase (C2139 Sigma), and 0.05 mg/ml DNase I (10104159001, Roche). The resulting suspension was filtered through a 70-μm cell strainer to obtain a single-cell preparation.

Flow cytometry staining of mouse samples

Staining: Cells were surface stained in DPBS at 4°C for 30 minutes with fixable viability dye eFluor 780, and combinations of the following listed antibodies (1:200, all from BioLegend; unless otherwise stated): CD11b BV510 (M1/70, 101263), CD45 BV650 (30-F11, 103151), CD80 BV605 (1:100, 16-10A1, 104729), Ly6G Percp-Cy5.5 (1A8, 127616), MHCII AF700 (M5/114.15.2, 107622), PD-L1 PE-Cy7 (1:100, 10F.9G2, 124314), Siglec-F BV421 (E50-2440, 552681 BD Biosciences). Samples were acquired on a LSRII Fortessa (BD Biosciences). Cells were surface stained in PBS at 4°C for 30 minutes with purified anti-mouse CD16/32 antibody (clone 93, BioLegend, Cat# 101301), Fixable Viability Dye 780 (eBioscience, Cat# 65-0865-14), CD11b (clone M1/70, BV510, BioLegend, Cat# 101263), Ly-6G (clone 1A8, PerCP/Cy5.5, BioLegend, Cat# 127615), Siglec-F (CD170; clone S17007L, FITC, BioLegend, Cat# 155503), CD62L (L-Selectin; clone MEL-14, BV605, BioLegend, Cat# 104437), CD8 (clone 53-6.7, APC, BioLegend, Cat# 100711), PD-1 (CD279; clone 29F.1A12, BV711, BioLegend, Cat# 135231), CD45 (clone 30-F11, BV650, BioLegend, Cat# 103151), TCR β (clone H57-597, PE-Cy7, BioLegend, Cat# 109221), NK1.1 (CD161; clone PK136, PE, BioLegend, Cat# 108707), CD4 (clone RM4-5, BV785, BioLegend, Cat# 100552), and CD44 (clone IM7, Alexa Fluor 700, BioLegend, Cat# 103025). For intracellular staining of Foxp3, cells were fixed and permeabilized according to the eBioscience™ Foxp3/Transcription Factor Staining Buffer Set (Invitrogen, Cat# 00-5523-00) followed by staining with anti-Foxp3 (clone MF-14, BV421, BioLegend, Cat# 126419). Samples were acquired in a LSRII Fortessa.

Data analysis: Analysis of flow cytometry data was performed using FlowJo Software. Relative cell frequencies, cell counts, or Mean Fluorescent Intensities (MFIs) were subsequently plotted in graphical formats using GraphPad Prism. Flow cytometry data was analyzed using FlowJo software. Relative cell frequencies and counts were plotted through graphical representations created in GraphPad Prism.

Statistical analysis: Statistical analyses were performed using GraphPad Prism. For statistical comparisons of two groups, two-tailed unpaired t-tests were used. For comparisons of more than two groups, a one-way analysis of variance (ANOVA) was used. Differences were considered statistically significant when $P < 0.05$. Statistical analyses were conducted using GraphPad Prism. Two-tailed unpaired t-tests were used to compare two groups, with statistical significance defined as $P < 0.05$.

Single-cell RNA sequencing of mouse neutrophils - experimental

CD45⁺ cell enrichment using magnetic beads: CD45 positive cells were enriched using anti-CD45 microbeads (130-052-301 Miltenyi Biotec) according to the manufacturer's instructions.

Cell hashing: Samples were pooled for shared scRNA-seq runs after labeling with sample tags (633793 BD Mouse Single-Cell Multiplexing Kit). Thereby, 10^6 cells from each sample were incubated in a staining buffer (DPBS, 1% BSA, 1% EDTA) and the respective sample tag for 20 min at room temperature ($20^\circ\text{C} \pm 2^\circ\text{C}$). After incubation, cells were washed twice with staining buffer before counting. From each sample 10-20.000 cells were pooled to have a total of around 60-80.000 cells for single cell capture.

Single cell capture, library preparation and sequencing: Single cell capture was done using the BD Rhapsody Express Single-Cell Analysis System (BD Biosciences) following the manufacturer's protocol. In short, cells were resuspended in 650 μL BD Sample Buffer supplemented with Rnase inhibitors (1:1.000 SUPERase 20 U μL^{-1} , AM2694 Thermo Fisher Scientific; NxGen RNase Inhibitor 40 U μL^{-1} , 30281-2 Lucigen) and loaded on a cartridge. To extract their RNA capture beads (v1 for tumor, disseminated and adjacent colon; enhanced beads for colon, blood and bone marrow) were added, followed by cell lysis. After single cell capture, libraries from the cDNA and the sample tags were prepared using the BD Rhapsody Whole Transcriptome Analysis Amplification Kit (633801 BD Biosciences). Libraries were then quantified using a Qubit Fluorometer (dsDNA HS kit: Q32851 Thermo Fisher Scientific) and a TapeStation 4200 (HS D500 types and reagents, 5067-5592 Agilent Technologies). Sequencing was performed on a NovaSeq6000 system (Illumina) with SP100 Reagents (100 cycles) and PhiX spike-in (20% for v1, 1% for enhanced beads) with a depth of around 20.000 reads/cell for WTA and 500-1000 reads/cell for the sample tags.

Single-cell RNA sequencing of mouse neutrophils - data analysis

Pre-processing, normalization, clustering and data integration: Demultiplexing was done using Bcl2fastq (v2.20.0433, Illumina). FASTQ files were then further processed using the BD Rhapsody™ WTA analysis pipeline (v1.12.1) on the Seven Bridge Genomics platform with default settings. Gene mapping was done with a STAR index generated using STAR⁴⁴ (v2.5.2b) with the gene code GRCm38 v25. R (v4.3) and Python (v3) were used for the downstream analysis. If not specified, the Seurat¹⁴¹ (v5.1.0) R package was used. After the first quality assessment, cells with fewer than 200 gene features, more than 5000 gene features and more than 25% of mitochondrial genes were removed from the dataset. Additionally, the R package decontX⁴⁵ (v1.0.0) was applied to adjust the dataset for cell free RNA. This approach uses a Bayesian model to estimate and remove cell free RNA contaminations. Data was log normalized and scaled with the Seurat functions 'NormalizeData' and 'ScaleData'. Thereby, the number of features, UMI counts and percentage of mitochondrial genes were regressed out. The top 2.000 variable feature genes were used for principal component analysis (PCA) followed by clustering using the Louvain algorithm with multi level refinements. The colon and tumor samples were processed separate from the blood and bone marrow samples. To account for batch effects in the colon and tumor samples the fast mutual nearest neighbors (FastMNN) algorithm (Seurat function 'FastMNNIntegration') was applied. The blood and bone marrow samples were merged without batch effect correction.

Annotation, extraction and integration of neutrophils and progenitors: Neutrophil clusters were identified using on the one hand neutrophil specific marker genes (S100a8, S100a9) and an automatic annotation approach called SingleR⁴⁶ (v2.4.1) with the murine ImmGen⁴⁷ (Immunological Genome Project) as reference which was loaded with the cellDex⁴⁶ (v1.12.0) R package. Neutrophil

clusters were then extracted and integrated from all tissues using the FastMNN approach mentioned before. Progenitors were identified with marker gene expression based on Garner et al.³⁵

Analysis of human neutrophil markers in mouse data: To analyze the expression of marker genes identified in human neutrophil clusters, the genes were first converted to mouse gene symbols using the orthogene Bioconductor package (v1.8.0) and then plotted in a heatmap using pheatmap (v1.0.12).

Pseudotime trajectory analysis with Slingshot: A pseudotime trajectory was inferred on the mnn.clusters of the combined mouse neutrophils dataset (bone marrow, blood and colon) using the slingshot⁴⁸ (v2.10.0) R package. Therefore, the Seurat object was first converted to a Single Cell Expression (SCE) object using sceasy R package (v0.0.7), which was then used for the slingshot analysis on the UMAP dimensional space. In brief, by minimum spanning tree (MST) construction using the 'getLineages' function, a global lineage structure was identified and pseudotime variables were inferred using the 'getCurves' function. To compare the pseudotime between tissue, the cellular distribution was visualized in a density plot across pseudotime. Transcriptional changes along the trajectory were analyzed using tradeSeq⁴⁹ (v1.16.0). Thereby, a negative binomial generalized additive model (NB-GAM) was applied to smooth the gene expression of the top 2000 variable features (Seurat function 'VariableFeatures') into the pseudotime lineage. This was done with the tradeSeq function 'fitGAM' for 5 knots. The knots represent points where smoothers are joined together. To extract DEGs between phenotypes (healthy and tumor) along the pseudotime the function 'conditionTest' was used on smoothened data using the default settings. Thereby a Wald statistics was applied with additional correction of p-values using false discovery rate (FDR). Selected genes were then plotted in a heatmap using pheatmap R package (v1.0.12).

Differential gene expression analysis: The Seurat function 'FindMarkers' with min.pct = 0.25 and logfc.threshold = 0.2 was used on normalized data to analyse DEGs between healthy and tumor bone marrow and blood. Thereby a non-parametric Wilcoxon Rank-Sum test was applied and p-values were corrected for multiple testing using the Bonferroni method.

Cell cycle phase scoring: Cells were grouped into S, G1 and G2/M phases based on expression of cell cycle specific genes from Tirosch et al.¹⁴² by applying the Seurat function 'CellCycleScoring'.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis was conducted using the statsmodels library in Python for scRNA-seq data or GraphPad Prism for flow cytometry and imaging data. Appropriate statistical tests, including linear models, t-tests, or Wilcoxon tests, were applied depending on the dataset and comparison. Single-cell data were aggregated into pseudobulk samples by biological replicate, considering only cell populations with ≥ 20 cells. P-values from untargeted analyses, such as differential gene expression, transcription factor activity, or pathway enrichment, were adjusted using FDR to account for multiple testing. For Shears analyses, the mean Shears coefficient per patient and cell type (≥ 20 cells) was visualized in boxplots weighted by $-\log_{10}(\text{p-value})$ to capture both effect size and statistical confidence. Boxes represent the interquartile range (IQR) with the median as the central line, and whiskers extend to $1.5 \times \text{IQR}$. Shears significance was evaluated using a Wilcoxon signed-rank test with FDR correction ($\text{FDR} < 0.05$). Detailed information on test types and significance levels for each analysis is provided in the figure legends.